

Relation Algebra

§1. Notation

Traditionally Adjunction $F \dashv G$ is defined as $F(x) \leq y$ iff $x \leq G(y)$, and you derive laws like this:

$$\begin{aligned} F(x) \leq F(x) &\implies x \leq G(F(x)) \implies F(x) \leq F(G(F(x))) \\ G(y) \leq G(y) &\implies F(G(y)) \leq y \implies G(F(G(y))) \leq G(y) \end{aligned}$$

By define $X : * \mapsto x$, $Y : * \mapsto y$ and use diagram order, application can be replaced by composition: $XF \leq Y$ iff $X \leq YG$

• write F as $/$, G as $\backslash$

• render $a \leq b$ as $\boxed{\begin{smallmatrix} a \\ b \end{smallmatrix}}$

• **rewrite** $X \leq XF$ as $1 \leq F$

(1a)

$$\boxed{\frac{X/}{Y}} = \boxed{\frac{X}{Y\backslash}}$$

adj

$$\begin{array}{c} \boxed{\frac{X/}{X/}} \xRightarrow{\text{adj}} \boxed{\frac{X}{X\wedge}} \xRightarrow{\text{rewrite}} \boxed{\frac{*}{\wedge}}_{\text{unit}} \begin{array}{l} \xRightarrow{\text{put } \backslash \text{ on left}} \boxed{\frac{\backslash}{\wedge}} \\ \xRightarrow{\text{put } / \text{ on right}} \boxed{\frac{/}{\vee}} \end{array} \\ \\ \boxed{\frac{Y\backslash}{Y\backslash}} \xRightarrow{\text{adj}} \boxed{\frac{Y\vee}{Y}} \xRightarrow{\text{rewrite}} \boxed{\frac{\vee}{*}}_{\text{counit}} \begin{array}{l} \xRightarrow{\text{put } / \text{ on left}} \boxed{\frac{\vee}{/}} \\ \xRightarrow{\text{put } \backslash \text{ on right}} \boxed{\frac{\wedge}{\backslash}} \end{array} \end{array}$$

now we have $\vee = /$ and $\wedge = \backslash$, and you just discovered string diagrams.

§1.1. Adjunctions

(1.1a)

$F \dashv G$	F 's type	monad FG	$1 \vDash FG$	$GF \vDash 1$	F preserves u	G preserves n	$FGF=F$	$GFG=G$
$\triangleleft \dashv \triangleleft$	$A \rightarrow A \otimes A$	$\triangleleft \triangleleft = 1$	$1 \vDash \triangleleft \triangleleft$	$\triangleleft \triangleleft \vDash 1$	$(RuS) \triangleleft = R \triangleleft uS \triangleleft$	$(RnS) \triangleright = R \triangleright nS \triangleright$	$\triangleleft \triangleleft \triangleleft = \triangleleft$	$\triangleright \triangleleft \triangleright = \triangleright$
$\multimap \dashv \multimap$	$A \rightarrow I$	$\multimap \multimap = T$	$1 \vDash \multimap \multimap$	$\multimap \multimap \vDash 1$	$(RuS) \multimap = R \multimap uS \multimap$	$(RnS) \multimap = R \multimap nS \multimap$	$\multimap \multimap \multimap = \multimap$	$\multimap \multimap \multimap = \multimap$
$\circ \dashv \circ$	$(A \rightarrow B) \rightarrow (B \rightarrow A)$	$\circ \circ = 1$	$R = R \circ \circ$	$R = R \circ \circ$	$(RuS) \circ = R \circ uS \circ$	$(RnS) \circ = R \circ nS \circ$	$R \circ \circ \circ = R \circ$	$R \circ \circ \circ = R \circ$
$\multimap \dashv \triangleleft \dashv \multimap$	$(X \otimes A \rightarrow Y) \rightarrow (X \rightarrow Y \otimes A)$	1	$(\multimap \triangleleft \otimes 1) (1 \otimes \multimap) = 1$	$(1 \otimes \multimap) (\triangleright \otimes 1) = 1$	$=$	$=$	$=$	$=$
$\triangleright \dashv \multimap \dashv \triangleleft$	$(X \rightarrow Y \otimes A) \rightarrow (X \otimes A \rightarrow Y)$	1	$(1 \otimes \multimap) (\triangleright \otimes 1) = 1$	$(\multimap \triangleleft \otimes 1) (1 \otimes \multimap) = 1$	$=$	$=$	$=$	$=$
$\Delta \dashv n$	$(A \rightarrow B) \rightarrow (A \rightarrow B)^2$	$R \mapsto RnR = R$	$R \vDash RnR$	$RnS \vDash R$	$\Delta(RuS) = \Delta Ru \Delta S$	$(RnT) n (SnU) = (RnS) n (TnU)$	$RnR = R$	$RnR = R$
$u \dashv \Delta$	$(A \rightarrow B)^2 \rightarrow (A \rightarrow B)$	$(R, S) \mapsto (RuS, RuS)$	$R \vDash RuS$	$RuR \vDash R$	$(RuT) u (SuU) = (RuS) u (TuU)$	$\Delta(RnS) = \Delta Rn \Delta S$	$RuR = R$	$RuR = R$
$\perp \dashv !$	$\{*\} \rightarrow (A \rightarrow B)$	—	—	$\perp \vDash R$	—	—	—	—
$\cdot S \dashv /S$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	S/S	$R \vDash (RS)/S$	$(T/S)S \vDash T$	$(RuT)S = RSuTS$	$(RnT)/S = R/SnT/S$	$((RS)/S)S = RS$	$((T/S)S)/S = T/S$
$S \cdot \dashv S \setminus$	$(B \rightarrow C) \rightarrow (A \rightarrow C)$	$S \setminus S$	$R \vDash S \setminus (SR)$	$S(S \setminus T) \vDash T$	$S(RuT) = SRuST$	$S \setminus (RnT) = S \setminus RnS \setminus T$	$S(S \setminus (SR)) = SR$	$S \setminus (S(S \setminus T)) = S \setminus T$
$Rn \dashv R \Rightarrow$	$(A \rightarrow B) \rightarrow (A \rightarrow B)$	$X \mapsto R \Rightarrow (XnR)$	$X \vDash R \Rightarrow (XnR)$	$Rn(R \Rightarrow Y) \vDash Y$	$Rn(XuY) = (RnX)u(RnY)$	$R \Rightarrow (XnY) = (R \Rightarrow X) n (R \Rightarrow Y)$	$Rn(R \Rightarrow (XnR)) = XnR$	$R \Rightarrow (Rn(R \Rightarrow Y)) = R \Rightarrow Y$
$\mathcal{D} \dashv \cdot T$	$(A \rightarrow B) \rightarrow \text{Cor } A$	$R \mapsto (\mathcal{D}R)T$	$R \vDash (\mathcal{D}R)T$	$\mathcal{D}(AT) \vDash A$	$\mathcal{D}(RuS) = \mathcal{D}Ru \mathcal{D}S$	$(AnB)T = ATnBT$	$\mathcal{D}((\mathcal{D}R)T) = \mathcal{D}R$	$(\mathcal{D}(AT))T = AT$
$\mathcal{R} \dashv T \cdot$	$(A \rightarrow B) \rightarrow \text{Cor } B$	$R \mapsto T(\mathcal{R}R)$	$R \vDash T(\mathcal{R}R)$	$\mathcal{R}(TA) \vDash A$	$\mathcal{R}(RuS) = \mathcal{R}Ru \mathcal{R}S$	$T(AnB) = TAnTB$	$\mathcal{R}(T(\mathcal{R}R)) = \mathcal{R}R$	$T(\mathcal{R}(TA)) = TA$
$\cdot f \dashv \cdot f^\circ$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	ff°	$1 \vDash ff^\circ$	$f^\circ f \vDash 1$	$(RuS)f = RfuSf$	$(RnS)f^\circ = Rf^\circ nSf^\circ$	$ff^\circ f = f$	$f^\circ ff^\circ = f^\circ$
$f^\circ \cdot \dashv f \cdot$	$(A \rightarrow C) \rightarrow (B \rightarrow C)$	ff°	$1 \vDash ff^\circ$	$f^\circ f \vDash 1$	$f^\circ(XuY) = f^\circ Xu f^\circ Y$	$f(XnY) = fXn fY$	$ff^\circ f = f$	$f^\circ ff^\circ = f^\circ$
$i \dashv E$	$\text{Map} \dashv \text{Rel}$	E	$\frac{1}{\exists} : A \rightarrow EA$	$\exists : EB \rightarrow B$	—	—	$\frac{\exists}{\exists} = 1$	$\frac{R \exists}{\exists} = R$
$\cdot \exists \dashv \frac{f \exists}{\exists}$	$\text{Map}(A, EB) \rightarrow (A \rightarrow B)$	1	$\frac{f \exists}{\exists} = f$	$\frac{R \exists}{\exists} = R$	—	—	$\frac{\frac{f \exists}{\exists}}{\exists} = f \exists$	$\frac{\frac{R \exists}{\exists}}{\exists} = \frac{R}{\exists}$
$\frac{\exists}{\exists} \dashv \cdot \exists$	$(A \rightarrow B) \rightarrow \text{Map}(A, EB)$	1	$\frac{R \exists}{\exists} = R$	$\frac{f \exists}{\exists} = f$	—	—	$\frac{R \exists}{\exists} = \frac{R}{\exists}$	$\frac{f \exists}{\exists} = f \exists$

§1.2. Composing adjunctions

(1.2a)

$$\begin{array}{c}
\boxed{\begin{array}{c} X \\ f(R/T) \end{array}} \xLeftrightarrow[f^\circ \cdot \dashv f \cdot] \boxed{\begin{array}{c} f^\circ X \\ R/T \end{array}} \xLeftrightarrow[\cdot T \dashv /T] \boxed{\begin{array}{c} (f^\circ X)T \\ R \end{array}} \\
\text{f a map} \\
\\
\xLeftrightarrow[\text{associativity}] \boxed{\begin{array}{c} f^\circ(XT) \\ R \end{array}} \xLeftrightarrow[f^\circ \cdot \dashv f \cdot] \boxed{\begin{array}{c} XT \\ fR \end{array}} \xLeftrightarrow[\cdot T \dashv /T] \boxed{\begin{array}{c} X \\ (fR)/T \end{array}} \\
\\
\xRightarrow[\text{indirect equality}] \boxed{\begin{array}{c} f(R/T) \\ (fR)/T \end{array}}
\end{array}$$

	$\cdot T$	$T \cdot$	$\cdot g$	$g \cdot$	Tn	$\circ$	$\circ \triangleleft$	Δ
$\cdot S$	$R/(ST) = (R/T)/S$	$T \setminus (R/S) = (T \setminus R)/S$	$R/(Sg) = (Rg^\circ)/S$	$g(R/S) = (gR)/S$	—	$(Y/S)^\circ = S^\circ \setminus (Y^\circ)$	$(RS)^\wedge = R^\wedge (S \otimes 1)$	$(T_1 n T_2)/S = T_1/S n T_2/S$
$S \cdot$	$S \setminus (R/T) = (S \setminus R)/T$	$(TS) \setminus R = S \setminus (T \setminus R)$	$S \setminus (Rg^\circ) = (S \setminus R)g^\circ$	$(g^\circ S) \setminus R = S \setminus (gR)$	—	$(S \setminus Y)^\circ = Y^\circ/S^\circ$	$((S \otimes 1)R)^\wedge = SR^\wedge$	$S \setminus (T_1 n T_2) = S \setminus T_1 n S \setminus T_2$
$\cdot f$	$R/(fT) = (R/T)f^\circ$	$T \setminus (Rf^\circ) = (T \setminus R)f^\circ$	$(fg)^\circ = g^\circ f^\circ$	—	—	$(fY)^\circ = Y^\circ f^\circ$	—	$(T_1 n T_2)f^\circ = T_1 f^\circ n T_2 f^\circ$
$f \cdot$	$f(R/T) = (fR)/T$	$(Tf^\circ) \setminus R = f(T \setminus R)$	—	$(fg)^\circ = g^\circ f^\circ$	—	$(fY)^\circ = Y^\circ f^\circ$	—	$f(T_1 n T_2) = fT_1 n fT_2$
Rn	—	—	—	—	$(RnT) \Rightarrow Y = R \Rightarrow (T \Rightarrow Y)$	$(R \Rightarrow Y)^\circ = R^\circ \Rightarrow (Y^\circ)$	—	$R \Rightarrow (T_1 n T_2) = (R \Rightarrow T_1) n (R \Rightarrow T_2)$
$\circ$	$(Y/T)^\circ = T^\circ \setminus (Y^\circ)$	$(T \setminus Y)^\circ = Y^\circ/T^\circ$	$(gY)^\circ = Y^\circ g^\circ$	$(gY)^\circ = Y^\circ g^\circ$	$(T \Rightarrow Y)^\circ = T^\circ \Rightarrow (Y^\circ)$	$Y^\circ \Rightarrow Y$	$(R^\circ)^\circ = (R^\circ \otimes 1)(1 \otimes \triangleright \circ)$ both bends: the $\circ$ section's display	$(T_1 n T_2)^\circ = T_1^\circ n T_2^\circ$
$\circ \triangleleft$	$(RT)^\wedge = R^\wedge (T \otimes 1)$	$((T \otimes 1)R)^\wedge = TR^\wedge$	—	—	—	$(R^\circ)^\circ = (R^\circ \otimes 1)(1 \otimes \triangleright \circ)$ both bends: the $\circ$ section's display	—	—
u	$(X_1 u X_2)T = X_1 T u X_2 T$	$T(X_1 u X_2) = TX_1 u TX_2$	$(X_1 u X_2)g = X_1 g u X_2 g$	$g^\circ(X_1 u X_2) = g^\circ X_1 u g^\circ X_2$	$Tn(X_1 u X_2) = (TnX_1)u(TnX_2)$	$(X_1 u X_2)^\circ = X_1^\circ u X_2^\circ$	—	—
$\perp$	$\perp T = \perp$	$T \perp = \perp$	$\perp g = \perp$	$g^\circ \perp = \perp$	$Tn \perp = \perp$	$\perp^\circ = \perp$	—	—

§1.3. Adjoint triples

Beyond $u \dashv \Delta \dashv n$, which §1.1 already carries as two rows, the table holds one adjoint triple with content: $\exists_f \dashv f^* \dashv \forall_f$ along a map f , once on each side of the composite. $f : A \rightarrow B$ throughout this subsection.

$$\begin{aligned} \cdot f \dashv \cdot f^\circ \dashv /f^\circ \\ f^\circ \cdot \dashv f \cdot \dashv f \setminus \end{aligned} \quad (1.3a)$$

The first two links of each chain are the map shunting rules, and the third is §1.1's division row read at a chosen divisor: $\cdot S \dashv /S$ at $S := f^\circ$, and $S \cdot \dashv S \setminus$ at $S := f$.

The third link is not the first over again. For a map f , $/f$ collapses to $\cdot f^\circ$; being a map is exactly what that collapse spends $R/(fS) = (R/S)f^\circ$. — f° is not a map, so $/f^\circ$ does not collapse in turn, and the three operators stay distinct. On $\text{Rel}(\mathbb{I}, A) = EA$ they are the image triple:

operator	acts by	name
$\cdot f$	$S \mapsto f[S]$	direct image, $\exists_f$
$\cdot f^\circ$	$T \mapsto f^{-1}[T]$	inverse image, f^*
$/f^\circ$	$S \mapsto \{b : f^{-1}(b) \subseteq S\}$	$\forall_f$

§1.1's $\mathcal{D} \dashv \cdot T$ is this same chain along the projection $A \otimes B \rightarrow A$, read through $\text{Rel}(A, B) = E(A \otimes B)$: $\mathcal{D}R = \{(a, a) : \exists b. aRb\}$, AT is the relation that ignores b altogether, and the third link is $R \mapsto 1 n R/T = \{(a, a) : \forall b. aRb\}$.

$$\mathcal{D} \dashv \cdot T \dashv 1 n \cdot /T \quad (1.3c)$$

Three is where it stops, and one map breaks both ends. Take $A = \{a_1, a_2\}$ and $B = \{b\}$: $f[\{a_1\}] n f[\{a_2\}] = \{b\}$ while $f[\{a_1\} n \{a_2\}] = \emptyset$, so $\cdot f$ does not preserve meets and has no left adjoint, and the same two subsets give $\forall_f(\{a_1\} u \{a_2\}) = \{b\}$ against $\forall_f\{a_1\} u \forall_f\{a_2\} = \emptyset$, so $/f^\circ$ does not preserve joins and has no right adjoint. A monic f restores the binary case and no more — τf still reaches only $\text{im } f$, and $\forall_f \perp = B \setminus \text{im } f$ is still not $\perp$. The chain extends only when f° is a map as well, and then $/f^\circ = \cdot f^\circ = \cdot f$ and it repeats forever. For an S with neither S nor S° a map there is no triple at all: $\cdot S \dashv /S$ is two links and stops.

The rows that chain forever say nothing by it: $\circ$ is an order-isomorphism of hom-posets, so it is adjoint to itself on both sides and $\circ \dashv \circ \dashv \circ$ has no end, and §1.1's other row of that kind, $\circ \triangleleft \dashv \triangleright \circ$, goes the same way.

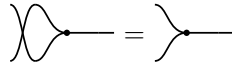
§2. Relations

$\mathbf{Rel}$ is a poset-enriched category with $(\otimes, \circ)$ where $\otimes$ is commutative cartesian product, and converse $\circ^{-1}$, and

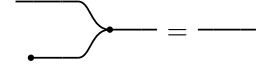
$\mathbf{C} : (\triangleright : A \otimes A \rightarrow A, \circ : I \rightarrow A)$ a commutative monoid with $\mathbf{C}^\circ : \mathbf{C}$. We write $\mathbf{C}^\circ$ as $(\triangleleft : A \rightarrow A \otimes A, \circ : A \rightarrow I)$



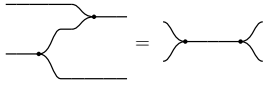
$\triangleright$ associative



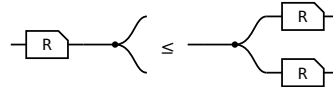
$\triangleright$ commutative



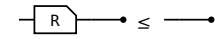
$\circ$ is its unit



Frobenius, one half — the other is its $\circ$



$R \triangleleft \leq \triangleleft (R \otimes R)$

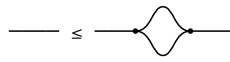


$R \circ \leq \circ$

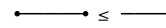
§2.1. $\forall$ object $\mathbf{A}$, $(\mathbf{A}, \triangleleft, \circ) \dashv (\mathbf{A}, \triangleright, \circ)$



$\triangleleft \leq 1$



$1 \leq \triangleleft$



$\circ \leq 1$



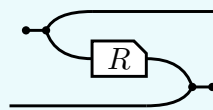
$1 \leq \circ$

(2.1a)

The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In $\mathbf{Rel}$ it reads $\{(a, a)\} \subseteq A \times A$ — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.

§3. $^\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ is a 2 functor

$^\circ$ is primitive, part of the data the first section lists. It turns both of R 's wires round, and the Frobenius structure DRAWS that — the picture and the formula below are R° , not its definition:



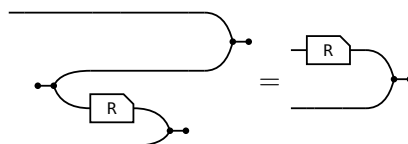
$$R^\circ = (\circ \leftarrow \triangleleft \otimes 1) (1 \otimes R \otimes 1) (1 \otimes \triangleright \rightarrow \circ)$$

where $\circ \leftarrow \triangleleft : I \rightarrow a \otimes a$ opens a pair of wires out of nothing and $\triangleright \rightarrow \circ : b \otimes b \rightarrow I$ closes one, so the input of R° is where the output of R was. Taking that formula as the definition would now be circular: $\triangleleft$ and $\rightarrow \circ$ are $\triangleright^\circ$ and $\circ \leftarrow$. The laws of $^\circ$ are that it is a contravariant 2-functor $^\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$:

$$(i) \ 1^\circ = 1 \quad (ii) \ (RS)^\circ = S^\circ R^\circ \quad (iii) \ (R \otimes S)^\circ = R^\circ \otimes S^\circ \quad (iv) \ R \leq S \text{ implies } R^\circ \leq S^\circ$$

§3.1. The slide

The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:



R° is DEFINED as the bending of $(R \otimes 1) \triangleright \rightarrow \circ$, so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the $\circ \leftarrow \triangleleft$ bends the a strand down and the $\triangleright \rightarrow \circ$ brings it back up, while the b strand rides through untouched. Nothing else is spent below.

§4. n is a commutative idempotent monoid on every hom-set

(4a)

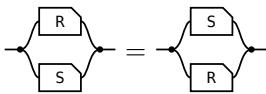
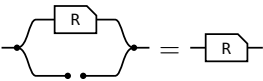
The **meet** of $R, S : \mathsf{a} \rightarrow \mathsf{b}$, the paper's **convolution**, is $R \mathsf{n} S := \langle (R \otimes S) \rangle$ — copy the input, run R and S on the two copies, merge the results — so what comes out is what both of them do.



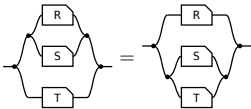
transcribed: a definition has no statement to export

On every hom-set it is associative, commutative and idempotent, with unit the maximal arrow $\mathsf{T} = \text{---} \circ \text{---}$ (the paper's Lemma 4.11).

(4b)



unit: one half of T per end — the merge's unit law absorbs the $\text{---} \circ \text{---}$, the copy's counit law the $\text{---} \circ \text{---}$ **commutative:** σ crosses $R \otimes S$ by naturality and is absorbed by cocommutativity and commutativity



associative: coassociativity and associativity; $\otimes$ re-brackets for nothing, **idempotent:** the one that is not bookkeeping — the lax copy law is the whole of it, worked in allegory2 being strict here

So $\leq$ is the order this monoid induces. $R \mathsf{n} S \leq R$ comes from the unit, and idempotency turns anything under both S and T into something under $S \mathsf{n} T$, since $R = R \mathsf{n} R \leq S \mathsf{n} T$.

And one law relating n to composition, which is **not** an equation:

(4c)

semi-distributivity, and what supplies it	picture
$R \ (S \mathsf{n} T) \sqsubseteq R S \mathsf{n} R T$ — the lax copy law. Equality exactly when R is single valued: the Maps section's F $(R \mathsf{n} S) = F R \mathsf{n} F S$.	

§5. Domain and range

The **domain** $\text{Dom } R \triangleq 1 \cap R R^\circ$ and the **range** $\text{Ran } R \triangleq \text{Dom } R^\circ$.



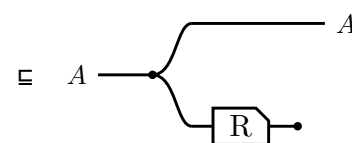
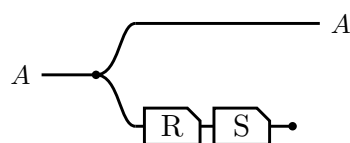
$\triangleright$ lands the return leg back on the value $\triangleleft$ handed out,
so $R^\circ \triangleright$ cuts to $\circ$ — Frobenius

Running R and throwing the result away leaves only the fact that R could fire, and $\text{Ran } R$ is the same picture with the box mirrored. In Rel both steps are $\{(a, a) : \exists b. a R b\}$.

$\text{Dom } R \triangleq 1 \cap R R^\circ$
$\text{Dom } R \subseteq A \iff R \subseteq AR$, for A coreflexive
$\text{Dom } (RS) \subseteq \text{Dom } R$
$\text{Dom } (R \cap S) = 1 \cap S R^\circ$
$R \text{ entire} \iff \text{Dom } R = 1 \iff 1 \subseteq R R^\circ$
$R \text{ simple} \iff R^\circ R \subseteq 1$
$R \text{ a map} \iff R \text{ entire and simple}$
$R, S \text{ entire} \implies RS \text{ entire}$ — likewise simple, likewise maps
$RS \text{ entire} \implies R \text{ entire}$

§5.1. Sliding the discard

$\text{Dom } (RS) \subseteq \text{Dom } R$, and a single glyph for Dom would have nothing to slide: with the box and the discard drawn apart, the law is one dot walking back along the lower strand.



$S^\circ \circ \circ$, the lax axiom for $\circ$ in the first section
— the discard slides back past S

Equality is S entire, which is the same picture read as $\text{Dom } R = 1 \iff R \text{ entire}$.

§6. Maps

surjective $\mathbb{1} \subseteq R^\circ R$, that is, R° entire.	single valued $R^\circ R \subseteq \mathbb{1}$
entire $\mathbb{1} \subseteq RR^\circ$. With single valued , a map .	injective $RR^\circ \subseteq \mathbb{1}$

(6a)

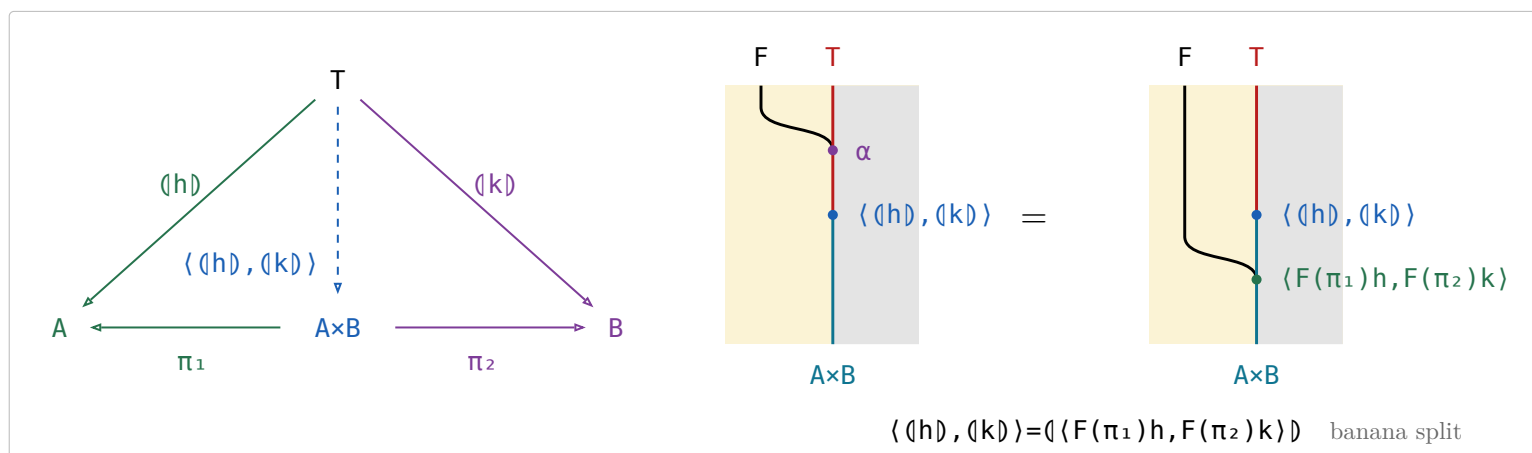
law $F (RnS) = FRnFS$ F single valued	picture
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(6b)

§7. Reduce in $\mathcal{S}et$

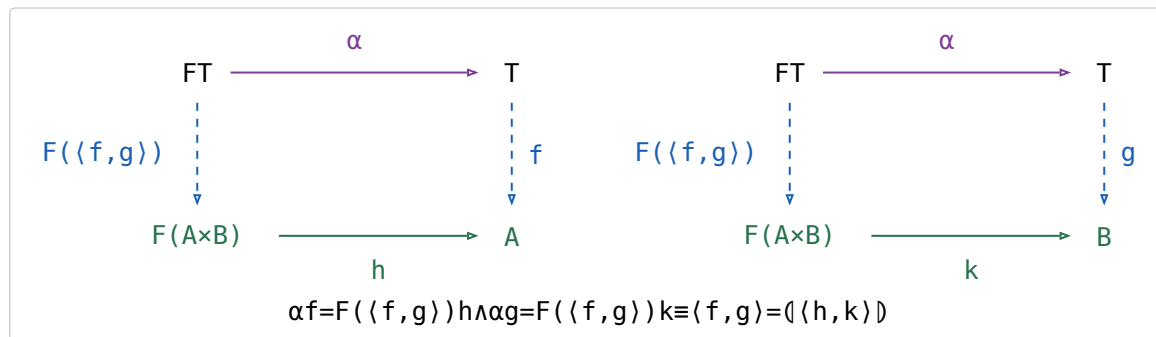
definition	type	note
<code>listr A ::= nil cons (A, listr A)</code>	<code>listr : Set → Set</code>	The cons-lists over A, the datatype every row below folds (B&dM p. 55).
<code>sum ≐ ([zero, plus])</code>	<code>sum : listr Nat → Nat</code>	<code>plus(a, b) = a + b</code> .
<code>length ≐ ([zero, π₂ succ])</code>	<code>length : listr A → Nat</code>	<code>π₂</code> drops the head and keeps the count of the tail, <code>succ</code> adds one for the head.
<code>average ≐ (sum, length) div</code>	<code>average : listr Nat → Real</code>	<code>div(m, n) = m/n</code> , with <code>div(0, 0) = 0</code> so <code>average</code> is total. Traverses the list twice.
banana-split law $\langle \langle h \rangle, \langle k \rangle \rangle = \langle \langle F(\pi_1)h, F(\pi_2)k \rangle \rangle$	$\langle \langle h \rangle, \langle k \rangle \rangle : T \rightarrow A \times B$	Any fork of folds is a single fold, hence one traversal — F the base functor.
what it reduces to $\alpha \langle \langle h \rangle, \langle k \rangle \rangle = F(\langle \langle h \rangle, \langle k \rangle \rangle) \langle F(\pi_1)h, F(\pi_2)k \rangle$	$\alpha \langle \langle h \rangle, \langle k \rangle \rangle : FT \rightarrow A \times B$	All that (11.6.1a) leaves to check: the fork satisfies the defining equation, and uniqueness finishes it.
the instance $\langle \text{sum}, \text{length} \rangle = ([\text{zeros}, \text{pluss}])$	$\langle \text{sum}, \text{length} \rangle : \text{listr Nat} \rightarrow \text{Nat} \times \text{Nat}$	<code>zeros = (zero, zero)</code> and <code>pluss(a, (b, n)) = (a + b, n + 1)</code> , so <code>average</code> runs in one pass.
<code>preds n = [n, n-1, ..., 1]</code>	<code>preds : Nat → [Nat]</code>	B&dM Ex 3.6 (p. 57), posed and not answered there: apply Ex 3.4 to write <code>preds</code> as $\langle k \rangle \pi_1$.

(7a)



§7.1. Fokkinga's mutual recursion theorem

- $F\text{-Alg}(\mathcal{A})$ — the algebras and their homomorphisms — has binary products whenever $\mathcal{A}$ does, and the forgetful $U : F\text{-Alg}(\mathcal{A}) \rightarrow \mathcal{A}$ creates them: the product of $h : FA \rightarrow A$ and $k : FB \rightarrow B$ is carried by $A \times B$, with structure $\langle F(\pi_1)h, F(\pi_2)k \rangle : F(A \times B) \rightarrow A \times B$.
- π_1 and π_2 are homomorphisms out of it, and $\langle p, q \rangle : X \rightarrow A \times B$ is a homomorphism **iff** p and q both are — substitute the structure and compare the two forks component by component.
- The banana-split law of (7a) IS that product's universal property read at the initial algebra: $\langle h \rangle$ and $\langle k \rangle$ are the unique homomorphisms to (A, h) and (B, k) , so their fork is the unique homomorphism into the product, which is $\langle \langle F(\pi_1)h, F(\pi_2)k \rangle \rangle$.
- Fokkinga's theorem is **strictly more general** and is not that product: in (7.1a) the two arrows leave $F(A \times B)$, not FA and FB , so each sees BOTH components. The product is the case where they factor as $F(\pi_1)h$ and $F(\pi_2)k$; B&dM's Ex 3.4 is the other special case (p. 58). What it says: **any algebra on a product carrier is folded by a fork**.



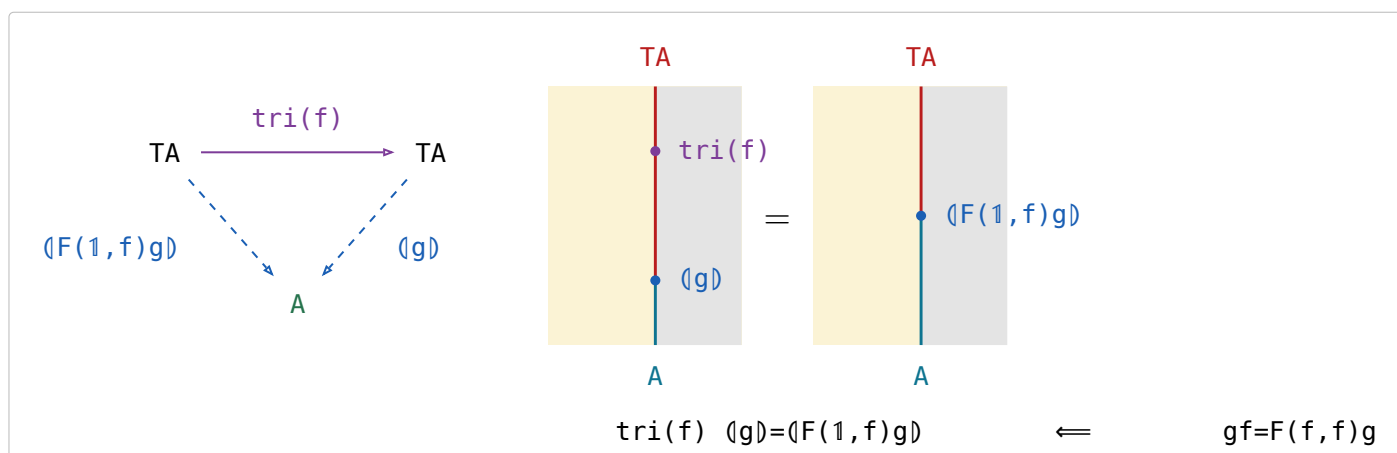
(7.1a)

§7.2. Ruby triangles

stage	definition
informally (p. 58)	$\text{tri}(f) [a_0, a_1, \dots, a_i, \dots, a_n] = [a_0, f a_1, \dots, f^i a_i, \dots, f^n a_n]$
for cons-lists (p. 59)	$\text{tri}(f) = \langle [\text{nil}, (1 \times \text{listr}(f)) \text{ cons}] \rangle$
with the base functor named	$F(A, B) = 1 + A \times B$, $\alpha = [\text{nil}, \text{cons}]$, so $\text{tri}(f) = \langle F(1, \text{listr}(f)) \alpha \rangle$
abstractly (p. 59)	F a bifunctor with initial type (α, T) : $\text{tri}(f) = \langle F(1, T(f)) \alpha \rangle$

(7.2a)

For the definition to make sense $f : A \rightarrow A$ is required, and then $\text{tri}(f) : TA \rightarrow TA$.



(7.2b)

(7.2b) is Horner's rule, generalised from cons-lists to any initial type; B&dM call it that because for cons-lists it is the schoolbook method for evaluating a polynomial (p. 58). Fusion reduces it to $F(1, T(f)) \alpha \langle g \rangle = F(1, \langle g \rangle) F(1, f) g$ (p. 59).

§7.3. Depth of a tree

the datatype	$\text{tree } A ::= \text{tip } A \mid \text{bin } (\text{tree } A, \text{tree } A)$, base functor $F(A,B)=A+B \times B$, initial type $([\text{tip}, \text{bin}], \text{tree})$	(7.3a)
the fold	$\llbracket g, h \rrbracket$ is the unique f with $f(\text{tip } a) = g\ a$ and $f(\text{bin } (x, y)) = h\ (f\ x, f\ y)$	
$\text{tree}(f)$	$\text{tree}(f) = \llbracket F(f, 1) \rrbracket [\text{tip}, \text{bin}]$; pointwise $\text{tree}(f)(\text{tip } a) = \text{tip } (f\ a)$ and $\text{tree}(f)(\text{bin } (x, y)) = \text{bin } (\text{tree}(f)\ x, \text{tree}(f)\ y)$	
max	$\text{max} = \llbracket 1, \text{bmax} \rrbracket$, where $\text{bmax } (a, b)$ is the larger of a and b	
depths	$\text{depths} = \text{tree}(\text{zero})\ \text{tri}(\text{succ})$ — replaces every tip by its depth in the tree, zero the constant function returning 0 and succ the successor	
depth	$\text{depth} = \text{depths}\ \text{max}$	

§8. / is all of

(8a)

$$x (R/S) y \iff \forall p. y S p \rightarrow x R p$$

$$x (S \setminus R) y \iff \forall p. p S x \rightarrow p R y$$

/compares images: $S(y) \subseteq R(x)$. \ compares preimages: $S^\circ(x) \subseteq R^\circ(y)$.

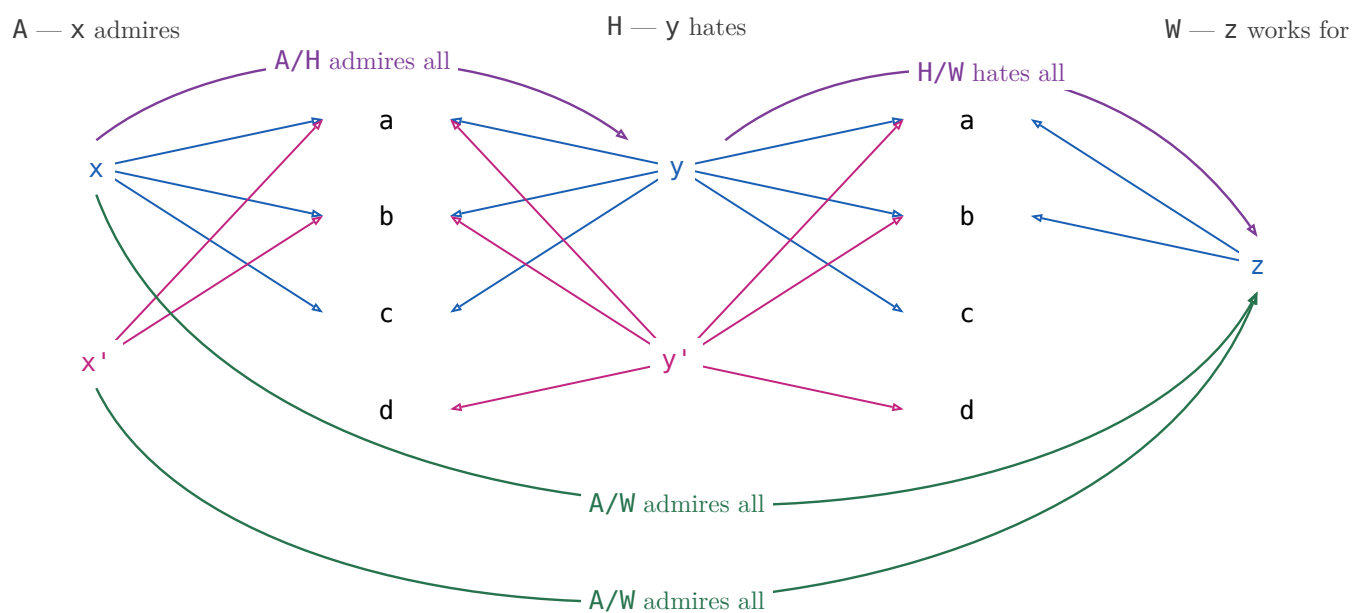
example: A admires, H hates, W works for

$x (A/H) y$ — x admires everyone y hates.

$x (H \setminus A) y$ — everyone who hates x admires y .

§8.1. $(R/S)(S/W) \sqsubseteq R/W$

(8.1a)



$$x (A/H) y \text{ — } x \text{ admires everyone } y \text{ hates}$$

$$y (H/W) z \text{ — } y \text{ hates everyone } z \text{ works for}$$

$$x (A/W) z \text{ — } x \text{ admires everyone } z \text{ works for}$$

(8.1b)

$(A/H)(H/W)$ is a path: $x \rightarrow y \rightarrow z$, and that is all of it. A/W also holds of x' , who admires everyone z works for — but x' does not admire everyone anybody hates, so nothing composes to it. The missing path is exactly the strictness of $(R/S)(S/W) \sqsubseteq R/W$.

$X \in R/S \iff X \subseteq R$ X is any X -to- Y pairing; one that only pairs X with a Y such that X admires everyone Y hates lies inside R/S , and R/S is the largest such.	
$X \in S \setminus R \iff S X \subseteq R$ The mirror — divide on the left when X comes first.	
$(R/S) S \subseteq R$ There is a Y such that X admires everyone Y hates, and p is one of the people Y hates — then X admires p too. Strict at $S=\emptyset$: R/S is everyone, $(R/S) S=\emptyset$.	
$S (S \setminus R) \subseteq R$ The mirror.	
associate: $R/(S_1 S_2) = (R/S_2)/S_1$ A friend's enemy is two hops: divide by the far end first.	
$(S_1 S_2) \setminus R = S_2 \setminus (S_1 \setminus R)$ The mirror.	
maps: $f (R/S) = (fR)/S$ Rename X before or after dividing — the licence to write fR/S .	
$R/(fS) = (R/S) f^\circ$ Rename Y : a map leaves a denominator as f° outside the box.	
$(R/S) (S/W) \subseteq R/W$ Someone who admires all of a hate-set that already covers everyone Z works for admires those people too.	
$1 \in R/R$ R/R runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only a and b admire each other's idols too, and still stay two people.	
$(R/R) (R/R) = R/R$ R/R is the preorder admires at least as much as , and a preorder is idempotent. Freyd writes $\sqsubseteq$; with $1 \in R/R$ above it is an equality.	
$(R/R) R = R$ Reaching p through someone whose idols X fully admires is reaching p directly, since X admires their own idols.	
$R/1 = R$ Dividing by 1 : p 's set is just $\{p\}$, so admiring all of it is admiring p .	
$R/(S_1 \cup S_2) = R/S_1 \cap R/S_2$ Admiring a combined hate-set is admiring each set in full.	
$S \setminus (R/W) = (S \setminus R)/W$ Which is why $S \setminus R/W$ needs no bracket.	

Fifteen laws, fifteen pictures, and not one shows a generator: η , υ , $\circ$ and composition are what the Frobenius generators build, and $/$ is none of those — it is posited, with nothing to unfold.

§9. $\frac{R}{S}$

$\frac{R}{S} \triangleq (R/S) \cap (S/R)^\circ$. In Rel x and y has the same image: $\forall p. (x R p \iff y S p)$

(9a)

$x (A/H) y$ — x admires everyone y hates

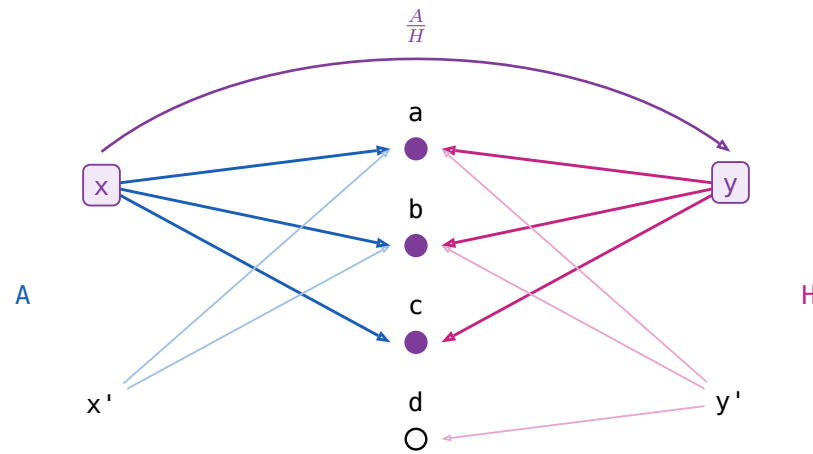
(9b)

$x ((H/A)^\circ) y$ — x admires only people y hates

$x ((A/H) \cap (H/A)^\circ) y$ — x admires only and all whom y hates

x admires exactly whom y hates: $/$ is **all of**, $\frac{R}{S}$ is **only all of**.

(9c)



law	the reading
$X \subseteq \frac{R}{S} \iff XS \subseteq R \text{ and } X^\circ R \subseteq S$	X may pair x with y only when x admires exactly whom y hates. Both halves must typecheck, so the operation is partial .
$(\frac{R}{S})^\circ = \frac{S}{R}$	Matching is symmetric.
$\frac{R}{S} \frac{S}{W} \subseteq \frac{R}{W}$	And transitive.
$\frac{R}{S} S \subseteq R$	$(\exists y. x(\frac{R}{S})y \wedge ySp) \rightarrow xRp$ x only admires whom y hates $\frac{R}{S} S = \text{Dom}(\frac{R}{S}) R$
$\frac{R}{R} R = R$	$(\exists y. x(\frac{R}{R})y \wedge yRp) \iff xRp$ x and y admire the same people $y=x$ always qualifies ($1 \in R \circ R$ below)
$1 \subseteq \frac{R}{R}$	$x(\frac{R}{R})y$ if x and y admires the same peoples.
$(\frac{R}{R})^2 = \frac{R}{R}$	So the relation admires the same people is an equivalence relation.
$X \subseteq \frac{R}{R} \iff XR \subseteq R$, for symmetric X	The largest symmetric arrow that leaves R alone.
$\frac{R}{1}$ is the simple part of R	The people who admire exactly one person and nobody else. It equals R only when R is simple, unlike $R/1=R$.
$\text{Dom } \frac{R}{S} = 1 \cap (R/S) (S/R)$	Its domain is the domain of simplicity of R .

(9d)

§10. Freyd's Power allegories

(10a)

A **power allegory** is a division allegory with one unary operation on arrows, $\exists$ **epsilon**, subject to

$$\begin{aligned} \exists_R \square &= R\square, & \exists_R &= \exists_R \square \\ \mathbb{1} &\subseteq (R/\exists_R)(\exists_R/R) & \exists_R &\text{ is } \mathbf{thick} \\ \exists_R &= \mathbb{1} & \exists_R &\text{ is } \mathbf{straight} \end{aligned}$$

$R\square$ is R 's target, an identity arrow. For $R : A \rightarrow B$ write $\exists : EA \rightarrow B$, dropping the subscript.

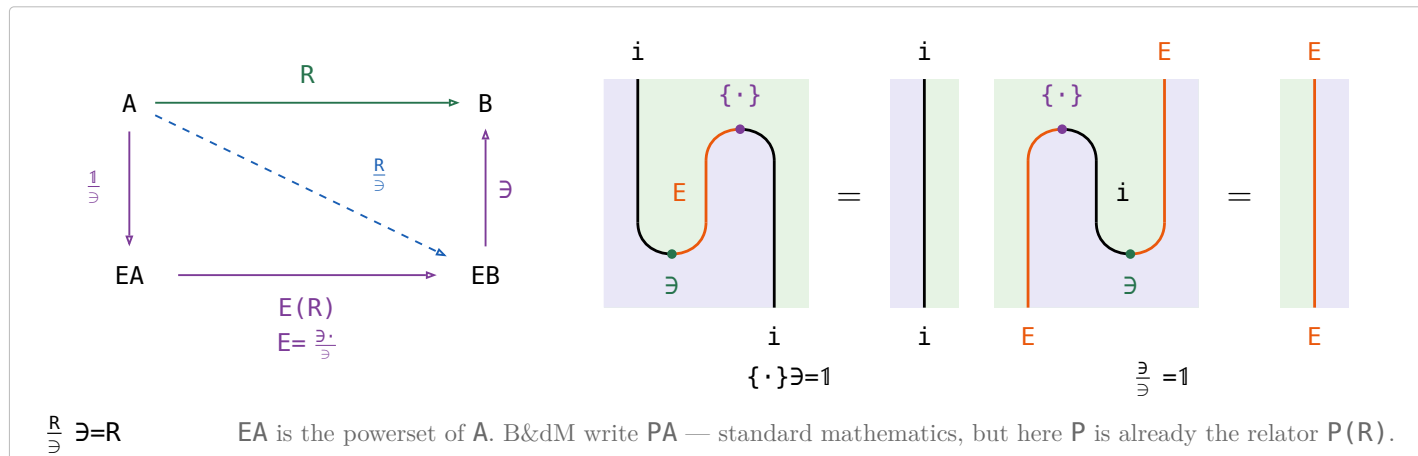
$$\exists \triangleq \exists^\circ : A \rightarrow EA$$

(10b)

law	the reading
1 $\exists_R \square = R\square, \exists_R = \exists_R \square$	One $\exists$ per object, not per arrow.
2 $\exists$ is thick	Comprehension: every x has a set of exactly the people x admires.
3 $\frac{R}{\exists} : A \rightarrow EB$, for $R : A \rightarrow B$	convert a relation to a function. $a \frac{R}{\exists} = \{b \mid a R b\}$
4 $\frac{R}{\exists}$ is a map	
5 $\frac{R}{\exists} \exists = R$	
6 $F \sqsubseteq \frac{F\exists}{\exists}$, F simple	A partial choice of sets is inside the total one.
7 $\frac{1}{\exists}$, the singleton map , monic	The one-person set.
8 $\frac{\exists}{\exists} = \mathbb{1}$	Make the set of a set, then read it back one level down.
9 fusion: $\frac{fR}{\exists} = f \frac{R}{\exists}$, f a map	Naturality of the unit, $f \frac{1}{\exists} = \frac{1}{\exists} (E(f))$.
10 $\frac{f}{\exists} = f \frac{1}{\exists}$, f a map	Rename first or take singletons first — the fusion row above at $R=1$.
11 $\frac{R}{S} = \frac{R}{\exists} \left(\frac{S}{\exists} \right)^\circ$	x and y match when R sends x and S sends y to the same set.
12 $E(R) \triangleq \frac{\exists R}{\exists}$, $E \triangleq \frac{\exists \cdot}{\exists}$	$E(R) : EA \rightarrow EB$, image of a set of A
13 $\frac{R}{\exists} = \frac{1}{\exists} E(R)$	$\frac{1}{\exists} : x \mapsto \{x\}$
14 $\frac{S}{\exists} E(R) = \frac{S}{\exists} \frac{\exists R}{\exists} = \frac{SR}{\exists}$	absorption — the monad's composition law, $\frac{S}{\exists} \diamond \frac{R}{\exists} = \frac{SR}{\exists}$, §10.2
15 $\text{subset} \triangleq \exists/\exists : EA \rightarrow EA$	$xs \text{ subset } ys \iff \forall a. ys \exists a \rightarrow xs \exists a$, that is $ys \subseteq xs$, not $xs \subseteq ys$.

§10.1. i -E Power Allegory defined as adjunction

(10.1a)



§10.2. $\mathcal{A} \cong \mathbf{Kleisli}(\mathbf{E})$

$$E(R) \triangleq \frac{\exists R}{\exists}, \{ \cdot \} = \frac{1}{\exists} : A \rightarrow EA, \text{ union} = E(\exists) = \frac{\exists \exists}{\exists} : E(EA) \rightarrow EA \quad (10.2a)$$

$$f \diamond g \triangleq f \circ E(g) \text{ union} : A \rightarrow EC, \text{ for } f : A \rightarrow EB \text{ and } g : B \rightarrow EC$$

the monad is on $\mathbf{Map}(\mathcal{A})$, not on the allegory: E is a relator on all relations, but $\{ \cdot \}$, **union** and $f \circ E(g)$ **union** are maps, and the Kleisli construction happens where they live.

$$\boxed{\frac{S}{\exists} \diamond \frac{R}{\exists}} \xrightarrow{\text{definition of } \diamond, \text{ union} = E(\exists)} \boxed{\frac{S}{\exists} E\left(\frac{R}{\exists}\right) E(\exists)} \xrightarrow{E \text{ a functor, } E(X)E(Y) = E(XY)} \boxed{\frac{S}{\exists} E\left(\frac{R}{\exists}\right)} \quad (10.2b)$$

$$\xrightarrow{(10b), \frac{R}{\exists} \exists = R} \boxed{\frac{S}{\exists} E(R)} \xrightarrow{(10b), \text{ absorption}} \boxed{\frac{SR}{\exists}}$$

the first three steps only unfold $\diamond$ — E is a functor, so $E\left(\frac{R}{\exists}\right)\text{union} = E\left(\frac{R}{\exists}\right)E(\exists) = E(R)$ and the **union** is gone.

Absorption, the row of (10b), is the whole law, and it is the functoriality of $\frac{\cdot}{\exists}$.

1 goes to $\frac{1}{\exists} = \{ \cdot \}$, the Kleisli identity, by definition — so $\frac{\cdot}{\exists}$ is an isomorphism of categories $\mathcal{A} \cong \mathbf{Kleisli}(\mathbf{E})$, and §10.1's $i \dashv E$ is the Kleisli adjunction. i is the identity on objects, so every object of the allegory is free.

§11. Relator

Every hom-set of an allegory is a poset, so an allegory is a **locally posetal 2-category**: the 2-cell from R to S IS $R \sqsubseteq S$. A **relator** $F : \mathcal{C} \rightarrow \mathcal{D}$ is a 2-functor between allegories:

$$F(1) = 1 \quad F(RS) = F(R)F(S) \quad R \sqsubseteq S \implies F(R) \sqsubseteq F(S)$$

Preserving $\circ$ is **not** asked for — $\circ$ is an identity-on-objects involution $\mathcal{C}^{\circ} \rightarrow \mathcal{C}$, no part of the 2-category.

the statement	B&dM	(11a)
For f a map, $F(f)$ is a map and $F(f^{\circ}) = F(f)^{\circ}$.	Lemma 5.1	
Over a tabular allegory a functor is a relator $\iff$ it preserves $\circ$.	Theorem 5.1	
$F(R^{\circ}) = F(R)^{\circ}$ for every R , so $F(R)^{\circ}$ needs no bracket.	after Theorem 5.1, p. 113	
Two relators agreeing on maps are equal.	Corollary 5.1	
$F(X \cap Y) = F(X) \cap F(Y)$ for X, Y coreflexive.	Ex 5.2	
$F(R \cap S) \sqsubseteq F(R) \cap F(S)$, and strictly.	Ex 5.2, the restriction	
$F(\text{Dom } R) = \text{Dom}(F(R))$ for F preserving $\circ$.	—	

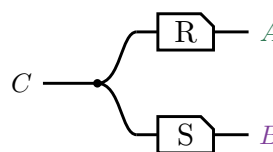
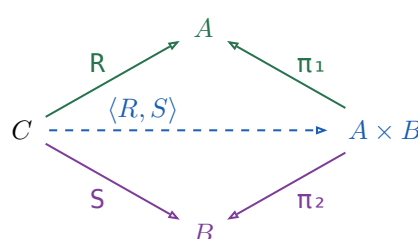
The **power relator** P — $xs \ P(R) \ ys \iff (\forall a \in xs. \exists b \in ys. a \ R \ b) \wedge (\forall b \in ys. \exists a \in xs. a \ R \ b)$ — is where the fourth is strict: for $R = \{(a_1, b_1), (a_2, b_2)\}$ and $S = \{(a_1, b_2), (a_2, b_1)\}$ the pair $(\{a_1, a_2\}, \{b_1, b_2\})$ is in $P(R) \cap P(S)$, while $R \cap S = \emptyset$.

§11.1. Fork $\langle R, S \rangle$

The **fork** of $R : C \rightarrow A$ and $S : C \rightarrow B$ is $\langle R, S \rangle \triangleq R \pi_1^{\circ} \cap S \pi_2^{\circ}$, where (π_1, π_2) is the tabulation of τ .

$$\langle R, S \rangle \pi_1 = (\text{Dom } S) R$$

$$\langle R, S \rangle \pi_2 = (\text{Dom } R) S$$



B&dM	the law	picture
(5.2)	$R \times S = \langle \pi_1 R, \pi_2 S \rangle$	
(5.3)	$\langle X, Y \rangle (R \times S) = \langle XR, YS \rangle$ both sides are the same strokes — (11.1.2a). Its $\sqsubseteq$ half is Ex 5.8, that half at $S := 1$ and at $R := 1$ B&dM's (5.4), (5.5) — stages of their proof of this row; Ex 5.6 is the $(R \times S) (U \times V) = (RU) \times (SV)$ it yields, at $R := 1$ and $V := 1$.	
(5.6)	$\langle R, S \rangle \pi_1 = (\text{Dom } S) R$ (11.1b)	
(5.7)	$\langle R, S \rangle \pi_2 = (\text{Dom } R) S$ (11.1b)	
(5.8)	$\langle X, Y \rangle \langle R, S \rangle^\circ = (XR^\circ) \cap (YS^\circ)$	
Ex 5.7	$\langle R, S \rangle^\circ \langle P, Q \rangle \sqsubseteq (R^\circ P) \times (S^\circ Q)$	
Ex 5.9	$f \langle R, S \rangle = \langle fR, fS \rangle$ f a map; it fails for an arbitrary arrow	
Ex 5.10	$F(R \times S) \text{unzip}(F) = \text{unzip}(F) (F(R) \times F(S))$ $\text{unzip}(F) \triangleq \langle F(\pi_1), F(\pi_2) \rangle$, a map	
Ex 5.11	$g = \text{curry}(f) \iff (g \times 1) \text{eval} = f$ reading $\times$ as the relational product, does Rel have exponentials?	

§11.2. Coproduct $[R, S] : A + B \rightarrow C$

(11.2a)

11.2.1a)

11.2.1b)



B&dM	the law	picture
(5.9)	$[R, S] = (\mathbf{l} \circ R) \cup (\mathbf{r} \circ S)$ (11.2a)'s first row	
(5.10)	$R+S = [R\mathbf{l}, S\mathbf{r}]$	
(5.11)	$[U, V] \circ [R, S] = (U \circ R) \cup (V \circ S)$ (11.2a)'s last row	
Ex 5.12	$X \triangleq [\mathbf{1}, \mathbf{1}] = \mathbf{l} \circ$ and $Y \triangleq [\mathbf{1}, \mathbf{1}] = \mathbf{r} \circ$, so $(X\mathbf{l}) \cup (Y\mathbf{r}) = [\mathbf{l}, \mathbf{r}] = \mathbf{1}$ which is (5.9)	
Ex 5.13	prove (5.11), and say why duality does not carry it over from the product law	
Ex 5.14	$(R+S) \cap ([P, Q][U, V] \circ)$ $= (R \cap (PU \circ)) + (S \cap (QV \circ))$ [P, Q][U, V] \circ is a full 2x2 of composites; R+S is diagonal, so the meet cuts the two off-diagonal branches	

§11.3. The power relator $P(R)$

(11.3a)

For $R : A \rightarrow B$,

$$P(R) \triangleq ((\exists R)/\exists) \cap ((\exists R^\circ)/\exists)^\circ : EA \rightarrow EB$$

$$E(R) \triangleq \frac{\exists R}{\exists} = ((\exists R)/\exists) \cap (\exists/(\exists R))^\circ$$

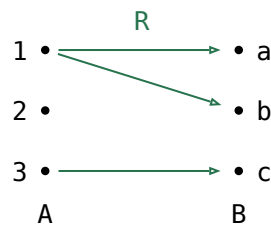
$$xs \ P(R) \ ys \iff (\forall a \in xs. \exists b \in ys. a \ R \ b) \wedge (\forall b \in ys. \exists a \in xs. a \ R \ b)$$

Every element of xs is related by R to some element of ys , and conversely.

(11.3b)

	in Rel	in words	$\text{im}(xs)$	partner
$(\exists R)/\exists$	$\forall y \in ys. \exists x \in xs. x \ R \ y$	$\forall y. \text{ some } xs \ R \ y$	$ys \subseteq \text{im}(xs)$	every y has a partner
$(\exists/(\exists R))^\circ$	$\forall y. (\exists x \in xs. x \ R \ y) \rightarrow y \in ys$	every y with some $xs \ R \ y$ is in ys	$ys \supseteq \text{im}(xs)$	ys leaves out no partner
$E(R)$	$ys = \{y \mid \exists x \in xs. x \ R \ y\}$	$ys = \text{every } y \text{ with some } xs \ R \ y$	$ys = \text{im}(xs)$	every y has a partner, and none is left out
$((\exists R^\circ)/\exists)^\circ$	$\forall x \in xs. \exists y \in ys. x \ R \ y$	$\forall x. x \ R \text{ some } ys$	—	every x has a partner
$P(R)$	$\forall y \in ys. \exists x \in xs. x \ R \ y$ and $\forall x \in xs. \exists y \in ys. x \ R \ y$	$\forall y. \text{ some } xs \ R \ y$ and $\forall x. x \ R \text{ some } ys$	—	every x and every y has a partner

(11.3c)



xs	$(\exists R)/\exists$	$(\exists/(\exists R))^\circ$	$E(R)$	$((\exists R^\circ)/\exists)^\circ$	$P(R)$
$\emptyset$	$\emptyset$	all 8 subsets of abc	$\emptyset$	all 8 subsets of abc	$\emptyset$
$\{1\}$	$\emptyset, a, b, ab$	ab, abc	ab	a, b, ab, ac, bc, abc	a, b, ab
$\{2\}$	$\emptyset$	all 8 subsets of abc	$\emptyset$	none	none
$\{3\}$	$\emptyset, c$	c, ac, bc, abc	c	c, ac, bc, abc	c
$\{1, 2\}$	$\emptyset, a, b, ab$	ab, abc	ab	none	none
$\{1, 3\}$	all 8 subsets of abc	abc	abc	ac, bc, abc	ac, bc, abc
$\{2, 3\}$	$\emptyset, c$	c, ac, bc, abc	c	none	none
$\{1, 2, 3\}$	all 8 subsets of abc	abc	abc	none	none

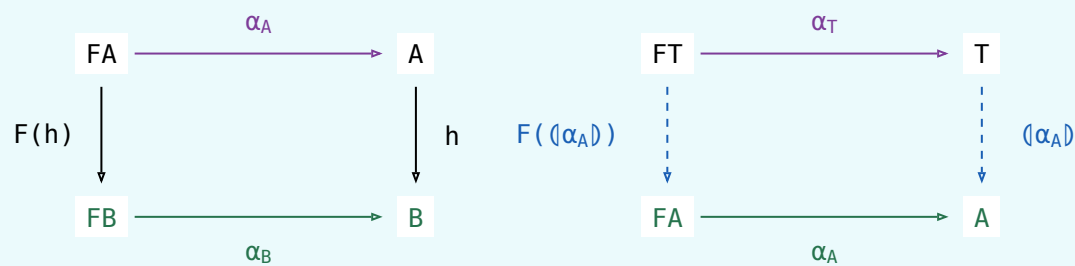
(11.3d)

law	the reading
$X \subseteq P(R) \iff X \subseteq \exists R \text{ and } X^\circ \subseteq \exists R^\circ$	One containment, and the same one at R° — which is the definition read off the two divisions. Hence $P(R^\circ) = P(R)^\circ$, and $R \subseteq S \implies P(R) \subseteq P(S)$.
$P(1) = \frac{\exists}{\exists} = 1$	The straightness axiom verbatim: extensionality is P 's unit law.
$P(f) = \frac{\exists f}{\exists}$, for f a map	In Rel, $xs \ P(f) \ ys \iff ys = \{f \ a \mid a \in xs\}$. The half at f° says every $a \in xs$ has its $f \ a$ on ys ; f has just the one image per a , so that already says ys contains everything xs reaches, which is the fraction's second half. For a map the two definitions coincide.
$P(RS) = P(R)P(S)$	$\sqsupseteq$ is the division cancellation laws. $\sqsubseteq$ is the one law in this section that is not a calculation: it needs a tabulation of $P(RS)$.

§11.4. Initial algebra

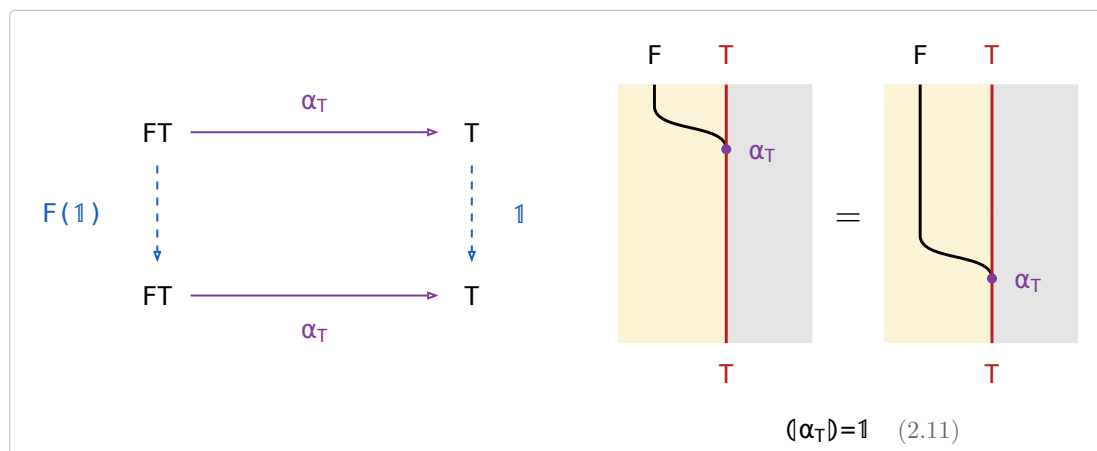
An **F-algebra** on A is a map $\alpha_A : FA \rightarrow A$. An **F-homomorphism** from α_A to α_B is a map $h : A \rightarrow B$ with $\alpha_B h = F(h) \alpha_A$. The **initial algebra** $\alpha_T : FT \rightarrow T$ is the F-algebra with exactly one F-homomorphism $\langle \alpha_A \rangle : T \rightarrow A$ to every F-algebra α_A .

(11.4a)



§11.4.1. Reflection

(11.4.1a)

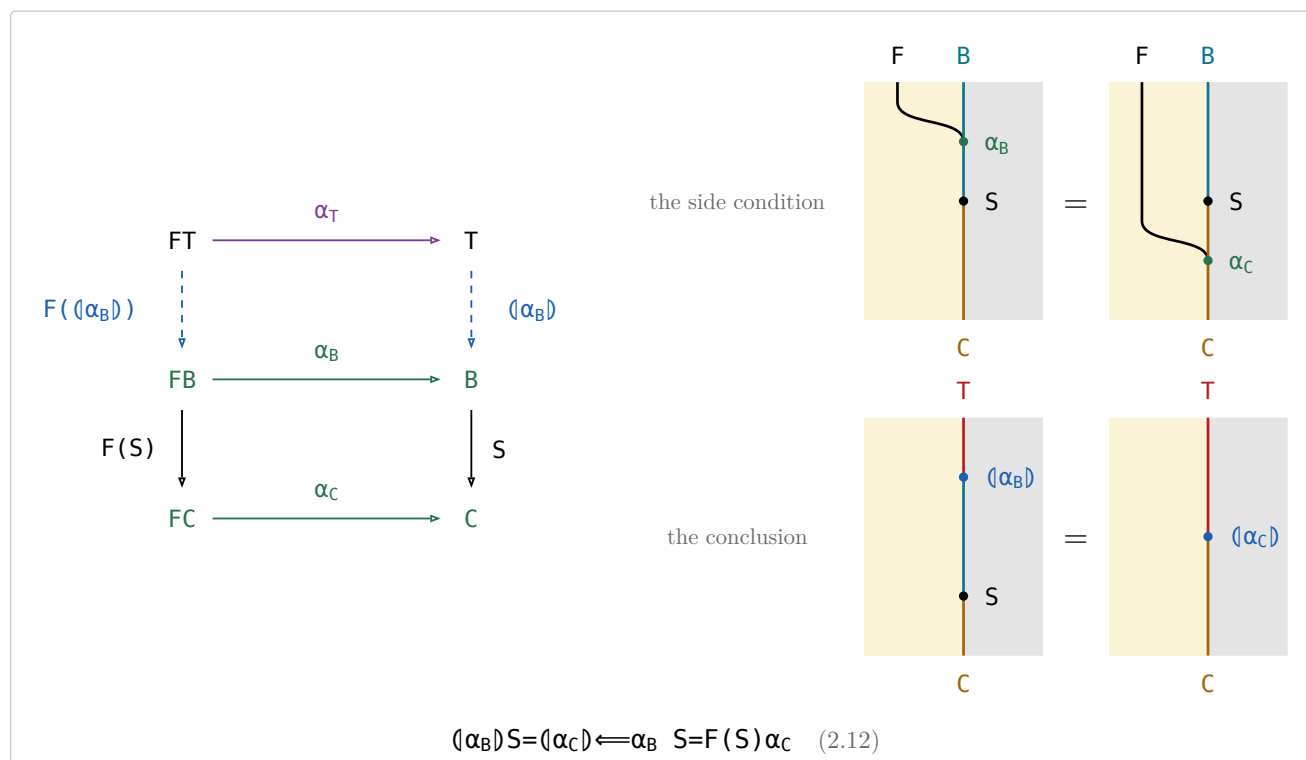


Taking a value apart with α_T and putting it straight back is doing nothing.

§11.4.2. Fusion

Fusion rewrites $\langle \alpha_B \rangle S$ through a second algebra $\alpha_C : FC \rightarrow C$ along an arrow $S : B \rightarrow C$.

(11.4.2a)

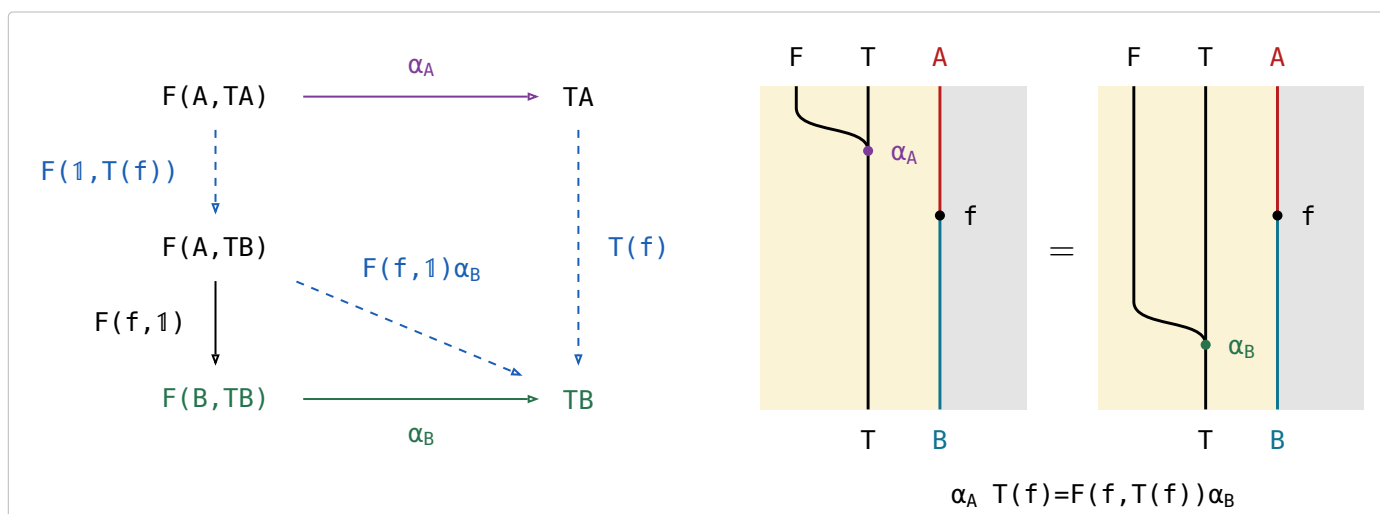


§11.4.3. Type functor

Let F be a bifunctor taking both the parameter A and the recursive position TA , with an initial algebra $\alpha_A : F(A, TA) \rightarrow TA$ for every object A . Then T is a functor, acting on a map $f : A \rightarrow B$ by

$$T(f) \triangleq (F(f, 1) \alpha_B) : TA \rightarrow TB$$

(11.4.3a)

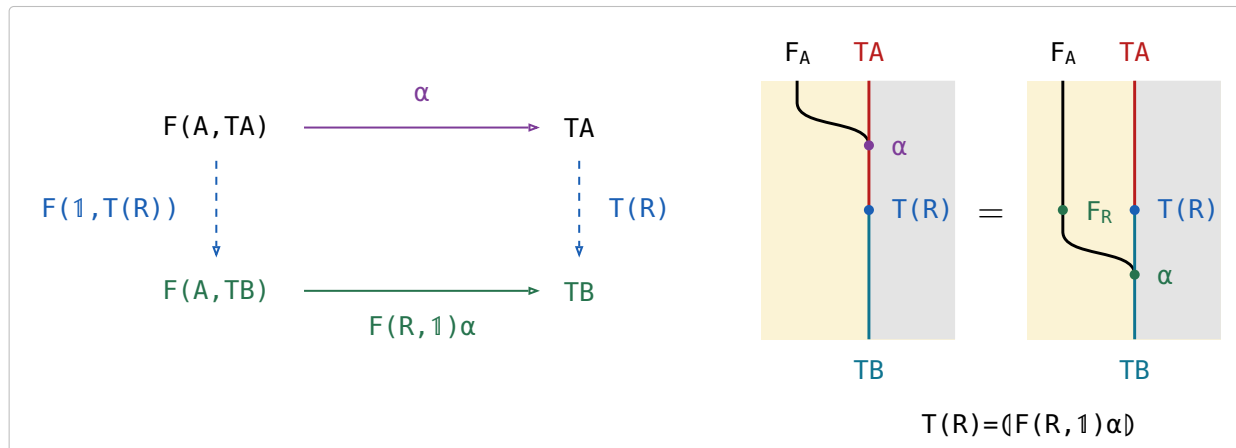


(11.4.3b)

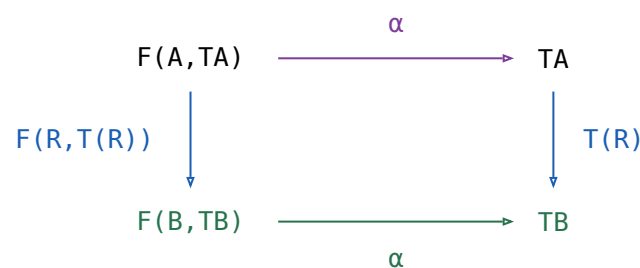
§11.5. Type relator

Let F be a binary relator with initial type (α, T) , so T is a type functor. $F(R, S)$ is its action on a pair, and $F(X)$ abbreviates $F(1, X)$, the F of the reduce section. For every object A the initial algebra is $\alpha : F(A, TA) \rightarrow TA$, among the maps. T acts on an arrow $R : A \rightarrow B$ by

$$T(R) = (F(R, 1)\alpha) : TA \rightarrow TB$$



name	law	what it says
the defining equation	$T(R) = (F(R, 1)\alpha)$	Rebuild the structure with α , applying R to the parameter on the way.
functor	$T(1) = 1$ and $T(R)T(S) = T(RS)$	Acting by the identity changes nothing, and two actions in a row are one action.
type functor fusion	$T(R)(Q) = (F(R, 1)Q)$	A relator action followed by a fold is a single fold — the intermediate structure is never built. The side condition holds because F is a bifunctor — $F(R, 1)F(1, (Q)) = F(R, (Q)) = F(1, (Q))F(R, 1)$.
naturality of α	$\alpha T(R) = F(R, T(R))\alpha$	Building and then mapping is the same as mapping the parts and then building, so α is natural from $G(R) = F(R, T(R))$ to T .
type relator	$T(R)^\circ = T(R^\circ)$	A datatype acts on relations, not only on maps — the map of the converse is the converse of the map.

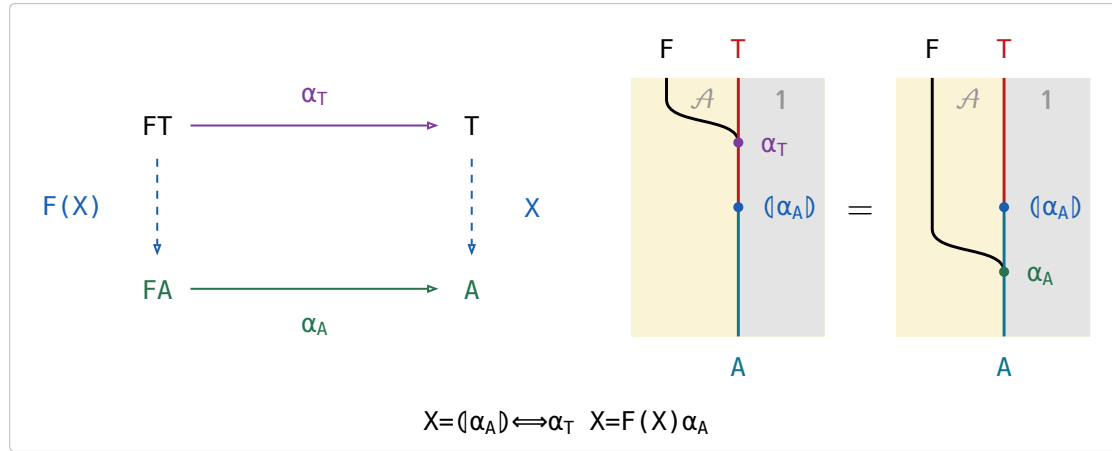


§11.6. Reduce

let F be a relator and has **initial algebra** $\alpha_T : FT \rightarrow T$ in the subcategory of functions. α_T is also initial in the allegory:

(11.6a)

§11.6.1. The defining equation



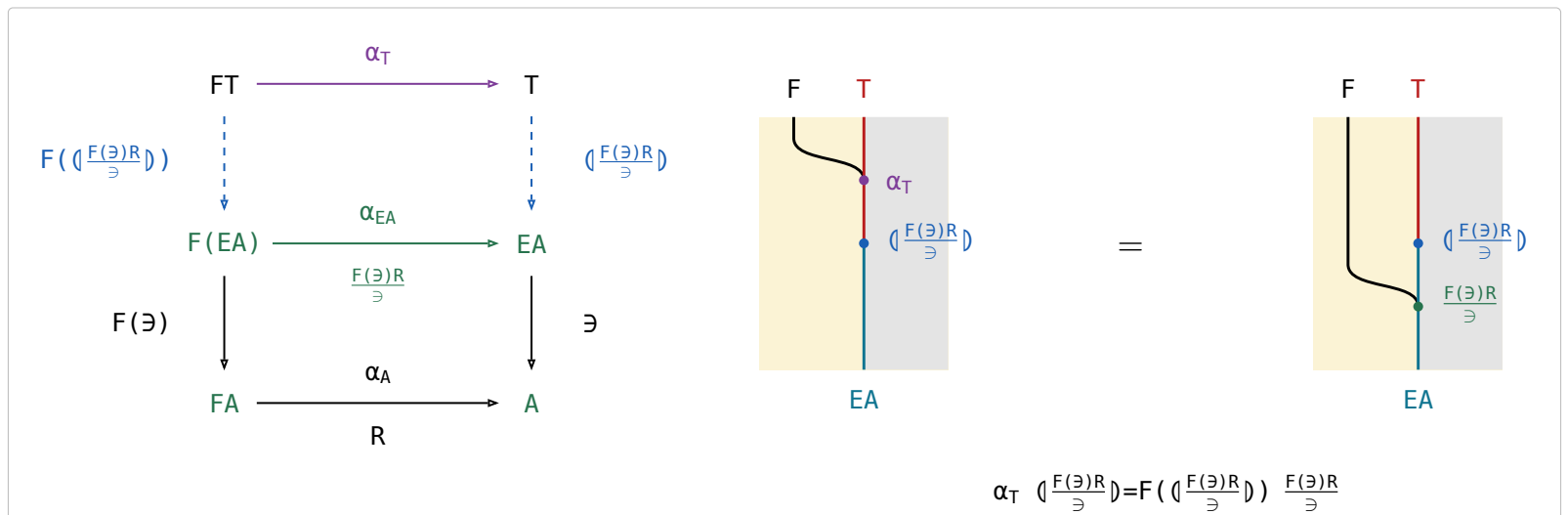
(11.6.1a)

An F -algebra is a **weakened** natural transformation. A transformation $F \Rightarrow 1$ on $\mathcal{A}$ would need a component $FX \rightarrow X$ at every object and a commuting square at every arrow, but F -Algebra only need it works on T and A .

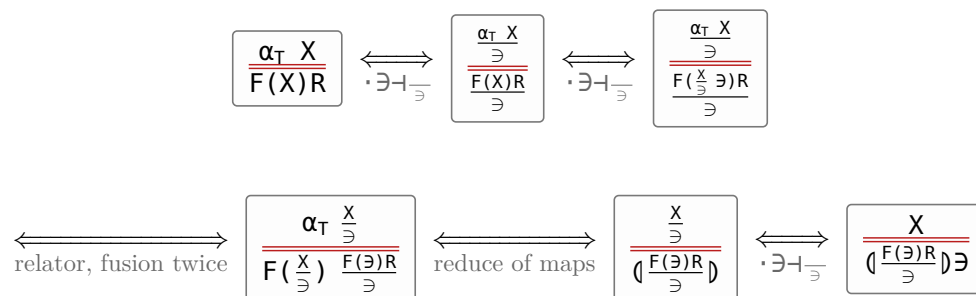
$\llbracket [c, f] \rrbracket = \text{reduce}(c, f)$		
part	Nat	[A]
datatype	$\text{Nat} ::= \text{zero} \mid \text{succ Nat}$	$[A] ::= \text{nil} \mid \text{cons } (A, [A])$
base functor F	$F(X) = 1 + X$	$F(X) = 1 + A \times X$
initial algebra α	$\alpha = [\text{zero}, \text{succ}]$ $: 1 + \text{Nat} \rightarrow \text{Nat}$	$\alpha = [\text{nil}, \text{cons}]$ $: 1 + A \times [A] \rightarrow [A]$
the fold, pointwise	$\llbracket [c, f] \rrbracket \text{ zero} = c$ $\llbracket [c, f] \rrbracket (\text{succ } n) = f (\llbracket [c, f] \rrbracket n)$	$\llbracket [c, f] \rrbracket \text{ nil} = c$ $\llbracket [c, f] \rrbracket (\text{cons } (a, x)) = f (a, \llbracket [c, f] \rrbracket x)$

(11.6.1b)

§11.6.2. $\llbracket R \rrbracket = (\llbracket \frac{F(\exists)R}{\exists} \rrbracket) \exists$



(11.6.2a)



(11.6.2b)

§12. Combinatorial functions

definition	type	note	(12a)
$[A] ::= \text{nil} \mid \text{cons } (A, [A])$	$\text{list} : A \rightarrow A$	B&dM's <code>list</code> under the short name it keeps from p. 125 on.	
$\text{list}(R) \triangleq ([\text{nil}, (R \otimes 1) \text{ cons}])$	$\text{list}(R) : [A] \rightarrow [B]$	The relator's action on $R : A \rightarrow B$: one R per element, the shape untouched.	
$\text{subseq} \triangleq ([\text{nil}, \text{cons} \pi_2])$	$\text{subseq} : [A] \rightarrow [A]$	xs subseq ys : ys is xs with elements dropped — <code>cons</code> keeps the head, π_2 drops it.	
$\text{prefix} \triangleq ([\text{nil}, \text{nilucons}])$ $= \text{cat}^\circ \pi_1 = \text{init}^*$	$\text{prefix} : [A] \rightarrow [A]$	ys is an initial segment of xs ; the first <code>nil</code> is where it stops early. $\text{init} \triangleq \text{snoc}^\circ \pi_1$.	
$\text{suffix} \triangleq \text{cat}^\circ \pi_2 = \text{tail}^*$	$\text{suffix} : [A] \rightarrow [A]$	The dual, $\text{tail} \triangleq \text{cons}^\circ \pi_2$; as a reduce it needs <code>snoc</code> -lists.	
$\text{segment} \triangleq \text{suffix prefix}$	$\text{segment} : [A] \rightarrow [A]$	A contiguous stretch of xs : a suffix, then a prefix of that.	
$\text{partition} \triangleq \text{concat}^\circ$	$\text{partition} : [A] \rightarrow [[A]]$	<code>cat</code> restricted to a non-empty first argument, so ys is a list of non-empty segments of xs .	
$\text{concat} \triangleq ([\text{nil}, \text{cat}])$	$\text{concat} : [[A]] \rightarrow [A]$	Joins the segments back up, which is why its converse splits a list.	
inits	$\text{inits} : [A] \rightarrow [[A]]$	Implements $\frac{\text{prefix}}{\ni}$, listing the prefixes by increasing length.	
tails	$\text{tails} : [A] \rightarrow [[A]]$	Implements $\frac{\text{suffix}}{\ni}$ by decreasing length — the opposite order, which is what <code>splits</code> $\triangleq (\text{inits}, \text{tails}) \text{ zip}$ needs to implement $\frac{\text{cat}^\circ}{\ni}$.	
$\text{filter}(p) \triangleq \frac{\text{subseq list}(p)}{\ni} \max R$	$\text{filter}(p) : [A] \rightarrow [A]$	The longest subsequence of xs whose every element passes p . p is a coreflexive and $\text{list}(p)$ the relator applied to it, so $\text{list}(p)$ is the test “every element passes” — no boolean anywhere. The longest is unique: only one subsequence keeps exactly the passing elements. The program is $([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2)])$. Ex 7.41; $\max R$ is (13.1a)	
$R \triangleq \text{length} \leq \text{length}^\circ$	$R : [A] \rightarrow [A]$	The preorder <code>filter</code> and <code>takewhile</code> maximise over: the longer list wins.	
$\text{takewhile}(p) \triangleq \frac{\text{prefix list}(p)}{\ni} \max R$	$\text{takewhile}(p) : [A] \rightarrow [A]$	The same with <code>prefix</code> for <code>subseq</code> : the longest prefix whose every element passes p . Unique, since the prefixes of one list are linearly ordered by length. Ex 7.39	
$\text{mss} \triangleq \frac{\text{segment sum}}{\ni} \max \leq$	$\text{mss} : [\text{Int}] \rightarrow \text{Int}$	Maximum segment sum. <code>segment</code> = <code>suffix prefix</code> splits it into $\frac{\text{prefix sum}}{\ni} \max \leq$ on each suffix, which the greedy theorem turns into a reduce. Ex 7.40	

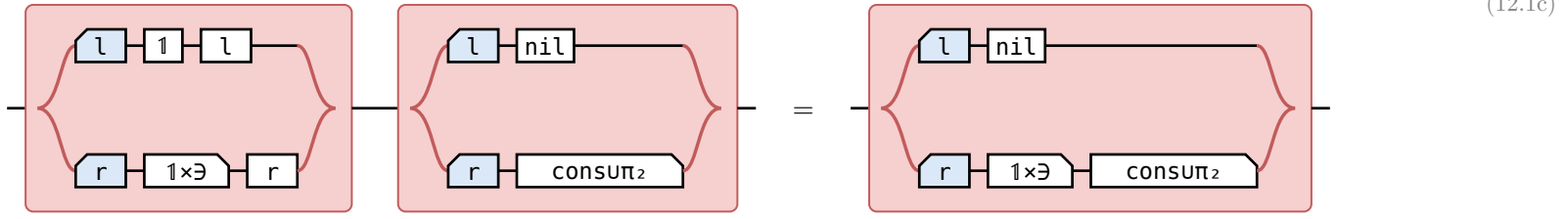
§12.1. $\frac{\text{subseq}}{\ni} = ([\text{nil}, \frac{1}{\ni}, (\frac{1 \times \ni}{\ni} \text{E}(\text{cons}), \pi_2) \text{ cup}])$

$$\text{cup} \triangleq \frac{\pi_1 \ni \cup \pi_2 \ni}{\ni} : EA \times EA \rightarrow EA, \text{ so } \frac{R \cup S}{\ni} = (\frac{R}{\ni}, \frac{S}{\ni}) \text{ cup}.$$

(12.1a)

$$\frac{F(\exists) \text{ [nil, cons}\pi_2]}{\exists} \xlongequal{\quad} FX=1+A \times X \xlongequal{\quad} \frac{(1+1 \times \exists) \text{ [nil, cons}\pi_2]}{\exists} \xlongequal{\text{coproduct}} \frac{\text{[nil, (1} \times \exists)(\text{cons}\pi_2)]}{\exists} \quad (12.1b)$$

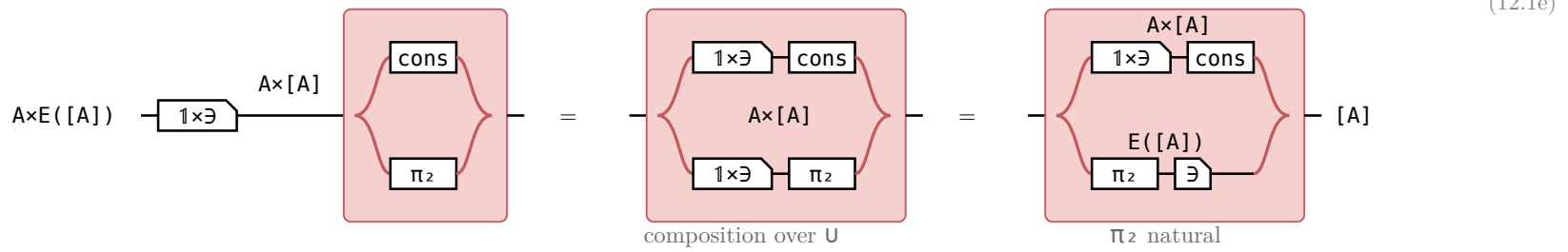
$$\xlongequal{\text{coproduct of maps}} \left[\frac{\text{nil}}{\exists}, \frac{(1 \times \exists)(\text{cons}\pi_2)}{\exists} \right] \xlongequal{\text{singleton}} \left[\text{nil} \frac{1}{\exists}, \frac{(1 \times \exists)(\text{cons}\pi_2)}{\exists} \right]$$



B&dM p. 124, “coproduct”: $R+S \triangleq [Rl, Sr]$ and $l[R, S]=R$, $r[R, S]=S$ — (11.2a).

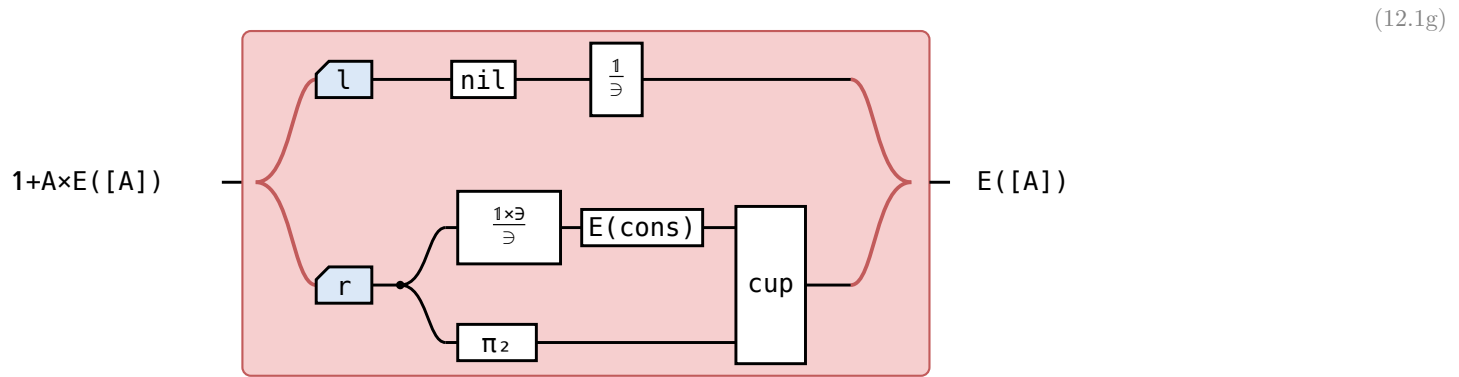
$$\frac{(1 \times \exists)(\text{cons}\pi_2)}{\exists} \xlongequal{\text{U composes, } \pi_2 \text{ natural}} \frac{(1 \times \exists) \text{ cons}\pi_2 \exists}{\exists} \xlongequal{\text{cup}} \left\langle \frac{(1 \times \exists) \text{ cons}}{\exists}, \frac{\pi_2 \exists}{\exists} \right\rangle \text{ cup} \quad (12.1d)$$

$$\xlongequal{\text{absorption, fusion}} \left\langle \frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right\rangle \text{ cup}$$



B&dM p. 124, “composition over join, naturality of π_2 ”: $T(X_1 \cup X_2) = TX_1 \cup TX_2$ — (1.2b); $(1 \times \exists) \pi_2 = \pi_2 \exists$ — (11.1.1b) at π_2 , an equality because 1 is entire.

$$\begin{array}{ccc} A \times E([A]) & \xrightarrow{\pi_2} & E([A]) \\ \downarrow 1 \times \exists & & \downarrow \exists \\ A \times [A] & \xrightarrow{\pi_2} & [A] \end{array}$$



B&dM §5.6, p. 124, which writes $P\text{cons}$; cons is a map, and there $P(\text{cons}) = E(\text{cons})$ — (11.3d).

§12.2. $\text{takewhile}(p) = ([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \infty \text{ nil})])$

$FX=1+A \times X$, $\alpha \triangleq [\text{nil}, \text{cons}]$, p a coreflexive on A , $R \triangleq \text{length} \leq \text{length}^\circ$, a preorder, $S \triangleq [\text{nil}, \infty \text{ nil} \cup (p \times 1) \text{ cons}] : F([A]) \rightarrow [A]$.

$\rightarrow$ nil is the constant nil, the second nil of $\text{prefix} = ([\text{nil}, \rightarrow \text{nilucons}])$; $(g \rightarrow X, Y)$ is X where g is defined and Y where it is not.

$\text{nil } R = \top$ nil is the shortest list, so it loses every $\max R$

$$\alpha \text{ prefix list}(p) \xrightarrow{\text{defining equation}} F(\text{prefix}) [\text{nil}, \rightarrow \text{nilucons}] \text{ list}(p) \quad (12.2b)$$

$$\text{list}(p) \text{ through cons} \xrightarrow{\quad} F(\text{prefix}) [\text{nil}, \rightarrow \text{nilu}(p \times \text{list}(p)) \text{ cons}]$$

$$\text{relator, prefix entire} \xrightarrow{\quad} [\text{nil}, \rightarrow \text{nilu}(p \times (\text{prefix list}(p))) \text{ cons}] \xrightarrow{\text{prefix list}(p) \text{ entire}} F(\text{prefix list}(p)) S$$

(11.6.1a) reads that off as $\text{prefix list}(p) = \{S\}$. (11.4.2a) cannot: $\text{list}(p)$ is not entire, $(1 \times \text{list}(p)) \rightarrow \text{nil} \subseteq \rightarrow \text{nil}$, and no algebra meets the side condition.

$$\begin{array}{c} (1 \times R) (\rightarrow \text{nilu}(p \times 1) \text{ cons}) \\ \xrightarrow{\text{relator, composition over } U} (1 \times R) \rightarrow \text{nilu}(p \times R) \text{ cons} \\ \xrightarrow{(5.1a), \subseteq (1 \times R) \text{ cons} \subseteq \text{cons } R} \rightarrow \text{nilu}(p \times 1) \text{ cons } R \\ \xrightarrow{R \text{ reflexive}} (\rightarrow \text{nilu}(p \times 1) \text{ cons}) R \end{array} \quad (12.2c)$$

the cons branch of $F(R) S \subseteq SR$; the nil branch is $\text{nil} \subseteq \text{nil } R$. $(1 \times R) \text{ cons} \subseteq \text{cons } R$ is $\text{cons length} = (1 \times \text{length}) \pi_2 \text{ succ}$ with succ monotone — a longer tail makes a longer list.

$$\begin{array}{c} \frac{S}{\exists} \max R \\ \xrightarrow{\text{coproduct of maps}} [\frac{\text{nil}}{\exists} \max R, \frac{\rightarrow \text{nilu}(p \times 1) \text{ cons}}{\exists} \max R] \\ \xrightarrow{\text{singleton, } R^\circ \text{ reflexive}} [\text{nil}, \frac{\rightarrow \text{nilu}(p \times 1) \text{ cons}}{\exists} \max R] \end{array} \quad (12.2d)$$

$$\text{nil } R = \top \xrightarrow{\quad} [\text{nil}, (\pi_1 p \rightarrow \text{cons}, \rightarrow \text{nil})]$$

the set is $\{\text{nil}\}$ where p fails on the head and $\{\text{nil}, \text{cons}(a, xs)\}$ where it holds, and nil loses the second — (13.1a) at a two-element set.

the law	what it says	(12.2e)
$(\frac{S}{\exists} \max R) \subseteq (\frac{S \cup}{\exists} \max R)$ Theorem 7.2, (13.3e) at R° ; $\max R = \min R^\circ$, so the hypothesis is monotonicity on $(R^\circ)^\circ = R$, which is (12.2c)	greedy: one longest p -prefix kept at each cons , instead of every p -prefix collected and one chosen at the end	
$\text{prefix}^\circ \text{ prefix} n R n R^\circ \subseteq 1$ context, (13.1b): only R between prefixes of one list counts	two prefixes of one list of equal length are equal, so $\frac{\text{prefix list}(p)}{\exists} \max R$ is simple — the longest, not a longest	
$\frac{\text{prefix list}(p)}{\exists} \max R$ entire	nil is always a p -prefix and R is connected on the prefixes of one list, so the longest exists	
$X \subseteq Y$, X entire, Y simple $\implies X = Y$ $([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \rightarrow \text{nil})])$ is a reduce of maps, hence entire	what turns the greedy $\subseteq$ into the heading's =	

§12.3. $\text{filter}(p) = ([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2)])$

F, α, p, R as in (12.2a); $S \triangleq [\text{nil}, \pi_2 \cup (p \times 1) \text{ cons}] : F([A]) \rightarrow [A]$, π_2 where (12.2a) has $\rightarrow \text{nil}$.

$1 \subseteq \pi_2 R \text{ cons}^\circ$ the tail is one shorter than the cons, so π_2 loses the $\max R$ at every step — where (12.2a)'s loser is nil

$$\alpha \text{ subseq list}(p) \xrightarrow{\text{defining equation}} F(\text{subseq}) [\text{nil}, \text{cons} \pi_2] \text{ list}(p) \xrightarrow{\text{list}(p) \text{ through cons, } \pi_2 \text{ natural}} F(\text{subseq}) [\text{nil}, (p \times \text{list}(p)) \text{ cons} \cup (1 \times \text{list}(p)) \pi_2] \quad (12.3b)$$

$$\text{relator, subseq list}(p) \text{ entire} \xrightarrow{\quad} F(\text{subseq list}(p)) S$$

(11.6.1a) reads that off as $\text{subseq list}(p) = \langle S \rangle$ — (12.2b)‘s chain with π_2 for $\rightarrow \text{nil}$. Here (11.4.2a) is blocked for the same reason, and π_2 is the branch that carries $\text{list}(p)$ out by naturality ((12.1f) at $\exists := \text{list}(p)$).

$$\begin{array}{c}
 \boxed{\begin{array}{c} (1 \times R) \\ (\pi_2 \cup (p \times 1)) \\ \text{cons} \end{array}} \xrightarrow[\pi_2 \text{ natural}]{\text{composition over } U,} \boxed{\begin{array}{c} \pi_2 R \cup (p \times R) \\ \text{cons} \end{array}} \xrightarrow[\text{branch}]{(12.2c) \sqsubseteq \text{cons}} \boxed{\begin{array}{c} \pi_2 R \cup (p \times 1) \\ \text{cons } R \end{array}} \xrightarrow[\text{composition over } U]{=} \boxed{\begin{array}{c} (\pi_2 \cup (p \times 1)) \\ \text{cons} \end{array}} R
 \end{array} \quad (12.3c)$$

$F(R)S \subseteq SR$, the nil branch again $\text{nil} \in \text{nil } R$. The first step is an equality: (12.2c) spends R reflexive on getting $\rightarrow \text{nil}$ past $1 \times R$, and π_2 needs only its naturality square.

$$\boxed{\frac{S}{\exists} \max R} \xrightarrow[(12.2d) \text{'s first two steps}]{=} \boxed{[\text{nil}, \frac{\pi_2 \cup (p \times 1)}{\exists} \text{cons} \max R]} \xrightarrow[1 \subseteq \pi_2 R \text{ cons}^\circ]{=} \boxed{[\text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2)]} \quad (12.3d)$$

at (a, xs) the set is $\{xs\}$ where p fails on a and $\{xs, \text{cons}(a, xs)\}$ where it holds, and xs loses the second — (13.1a) at a two-element set. The head is dropped, not the whole tail: that is the one place π_2 shows against (12.2d)‘s $\rightarrow \text{nil}$.

the law	what it says
$\langle \frac{S}{\exists} \max R \rangle \subseteq \langle \frac{S \rangle}{\exists} \max R$ (12.2e)‘s first row at this S , its condition (12.3c)	greedy: one longest p -subsequence kept at each cons
$S^\circ S n R n R^\circ \subseteq 1$ fails at $S := \text{subseq list}(p)$ (12.2e)‘s uniqueness row does not transfer	two p -subsequences of one list can have equal length and differ: the prefixes of one list are linearly ordered by length, its subsequences are not
$\frac{\text{subseq list}(p)}{\exists} \max R$ simple (12a)	a p -subsequence that drops a passing element is beaten by the one that keeps it, so only the subsequence keeping exactly the passing elements survives $\max R$ — the longest, not a longest
$X \subseteq Y$, X entire, Y simple $\implies X = Y$ (12.2e)‘s last row at this S	what turns the greedy $\subseteq$ into the heading’s $=$; $\langle [\text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2)] \rangle$ is again a reduce of maps, hence entire

§12.4. $\text{mss} = \langle [\text{zero wrap}, ((1 \times \text{head}) \oplus, \pi_2) \text{cons}] \rangle \max \leq$

$FX = 1 + A \times X$, $A := \text{Int}$, $\alpha \triangleq [\text{nil}, \text{cons}]$, $\text{sum} = \langle [\text{zero}, \text{plus}] \rangle$ and $\text{segment} = \text{suffix prefix}$ from (7a) and (12a).

$\text{head} \triangleq \text{cons}^\circ \pi_1$, $\text{wrap} \triangleq (1, \rightarrow \text{nil}) \text{cons}$ the head of a list and the one-element list, beside (12a)‘s $\text{tail} \triangleq \text{cons}^\circ \pi_2$

$\oplus \triangleq \frac{\rightarrow \text{zerouplus}}{\exists} \max \leq$ B&dM’s $\text{oplus} = \max \wedge (\text{zerouplus})$; the set at (a, b) is $\{0, a+b\}$, so $\oplus$ is the larger of the two

$$\begin{array}{c}
 \boxed{\frac{\text{segment sum}}{\exists} \max \leq} \xrightarrow[\text{segment} = \text{suffix prefix}]{=} \boxed{\frac{\text{suffix (prefix sum)}}{\exists} \max \leq} \xrightarrow[\text{absorption}]{=} \boxed{\frac{\text{suffix}}{\exists} E(\text{prefix sum}) \max \leq} \\
 \xrightarrow[\frac{R}{\exists} \exists = R, \text{union} = E(\exists)]{=} \boxed{\frac{\text{suffix}}{\exists} E(\frac{\text{prefix sum}}{\exists}) \text{union} \max \leq} \xrightarrow[(13.1b), \text{the sets non-empty}]{=} \boxed{\frac{\text{suffix}}{\exists} E(\frac{\text{prefix sum}}{\exists}) E(\max \leq) \max \leq} \\
 \xrightarrow[\text{relator}]{=} \boxed{\frac{\text{suffix}}{\exists} E(\frac{\text{prefix sum}}{\exists} \max \leq) \max \leq}
 \end{array} \quad (12.4b)$$

B&dM’s $\max P(\max \wedge (\text{sum prefix})) \wedge \text{suffix}$, mirrored. The union step is (13.1b)‘s $P(\min R) \min R = P(\text{Dom}(\min R)) \text{union} \min R$ — every suffix has the empty prefix, so Dom is 1 here — and $P(f) = E(f)$ at the map it is applied to ((11.3d)).

$$\begin{array}{c}
 \boxed{\begin{array}{c} [\text{nil}, \rightarrow \text{nilucons}] \\ \text{sum} \end{array}} \xrightarrow[\text{composition over } U]{\text{coproduct of maps,}} \boxed{\begin{array}{c} [\text{nil sum}, \rightarrow \text{nil}] \\ \text{sumucons sum} \end{array}} \xrightarrow[\text{sum's defining equation}]{=} \boxed{\begin{array}{c} [\text{zero}, \rightarrow \\ \text{zerou}(1 \times \text{sum}) \text{plus}] \end{array}} \\
 \xrightarrow[(1 \times \text{sum}) \rightarrow \rightarrow \rightarrow, \text{sum entire}]{=} \boxed{[\text{zero}, (1 \times \text{sum})(\rightarrow \text{zerouplus})]} \xrightarrow[\text{relator}]{=} \boxed{F(\text{sum}) [\text{zero}, \rightarrow \text{zerouplus}]}
 \end{array} \quad (12.4c)$$

(11.4.2a) at $\alpha_B := [\text{nil}, \multimap \text{nilucons}]$, $S := \text{sum}$: the side condition, so $\text{prefix sum} = ([\text{zero}, \multimap \text{zerouplus}])$. prefix is the reduce, sum the map fused into it — the intermediate list is gone.

$$\begin{array}{c}
 \boxed{(1 \times \leq^\circ) (\multimap \text{zerouplus})} \xrightarrow[\text{over } U]{\text{relator, composition}} \boxed{(1 \times \leq^\circ) \multimap \text{zerou}(1 \times \leq^\circ) \text{ plus}} \xrightarrow[(5.1a), (\leq^\circ \times \leq^\circ)]{\sqsubseteq} \boxed{\multimap \text{zerouplus} \leq^\circ} \xrightarrow[\text{reflexive}]{\leq^\circ} \boxed{(\multimap \text{zerouplus}) \leq^\circ}
 \end{array} \quad (12.4d)$$

the **plus** branch of $F(\leq^\circ) S \sqsubseteq S \leq^\circ$; the **zero** branch is $\text{zero} \sqsubseteq \text{zeros} \leq^\circ$. $(\leq \times \leq) \text{ plus} \sqsubseteq \text{plus} \leq$ is (13.3a), written $+$ there, and (13.3e)'s first row carries it to $\leq^\circ$ — **plus** is a map, so it is monotonic on an order and on its opposite together.

$$\boxed{\frac{[\text{zero}, \multimap \text{zerouplus}]}{\ni} \text{ max} \leq} \xrightarrow[\text{coproduct of maps}]{=} \boxed{[\frac{\text{zero}}{\ni} \text{ max} \leq, \frac{\multimap \text{zerouplus}}{\ni} \text{ max} \leq]} \xrightarrow[\text{singleton, } \leq^\circ \text{ reflexive}]{=} \boxed{[\text{zero}, \oplus]} \quad (12.4e)$$

with (12.4d) the greedy theorem gives $([\text{zero}, \oplus]) \sqsubseteq \frac{\text{prefix sum}}{\ni} \text{ max} \leq$ — B&dM's own containment — and (12.4g)'s second row makes it an equality.

$$\boxed{[\text{nil wrap}, ((1 \times \text{head}) \text{ cons}, \pi_2) \text{ cons}] \text{ list}(g)} \xrightarrow[\text{coproduct of maps, wrap and cons natural}]{=} \boxed{[\text{nil } g \text{ wrap}, ((1 \times \text{head}) \text{ cons } g, \pi_2) \text{ cons}] \text{ list}(g)} \quad (12.4f)$$

$$\xrightarrow[\text{g's defining equation, list}(g) \text{ head} = \text{head } g]{=} \boxed{[c \text{ wrap}, ((1 \times (\text{list}(g) \text{ head})) f, (1 \times \text{list}(g)) \pi_2) \text{ cons}]} \xrightarrow[\text{fork, relator}]{=} \boxed{F(\text{list}(g)) [c \text{ wrap}, ((1 \times \text{head}) f, \pi_2) \text{ cons}]}$$

$g \triangleq ([c, f])$ and $\text{tails} = ([\text{nil wrap}, ((1 \times \text{head}) \text{ cons}, \pi_2) \text{ cons}])$, since $\text{tails}(\text{cons}(a, xs))$ is $\text{cons}(\text{cons}(a, \text{head}(\text{tails } xs)), \text{tails } xs)$. (11.4.2a) then reads off $\text{tails list}(g) = ([c \text{ wrap}, ((1 \times \text{head}) f, \pi_2) \text{ cons}])$, at $c, f := \text{zero}, \oplus$ the heading's fold.

the law	what it says	(12.4g)
$(\frac{S}{\ni} \text{ max} \leq) \sqsubseteq \frac{(\frac{S}{\ni})}{\ni} \text{ max} \leq$ (13.3e) at $\leq^\circ$, $S := [\text{zero}, \multimap \text{zerouplus}]$, condition (12.4d)	greedy: one running maximum kept at each cons , instead of every prefix sum collected and one chosen at the end	
$X \sqsubseteq Y$, X entire, Y simple $\implies X = Y$ (12.2e)'s last row at this S	$([\text{zero}, \oplus])$ is a reduce of maps, and the prefix sums of one list are finitely many integers with one greatest, so (12.4e)'s $\sqsubseteq$ is an equality	
$\text{list}(g) \text{ head} = \text{head } g$ the one step of (12.4f) the note has no law for	head is a partial map, undefined at nil ; the equality holds because tails never returns an empty list, which is a fact about tails , not a naturality square	
tails implements $\frac{\text{suffix}}{\ni}$, $\text{list}(f)$ implements $E(f)$ (12a)	what makes (12.4b) a program: one fold builds the $n+1$ running maxima, and the final $\text{max} \leq$ reads them in one more pass, so mss is linear	

§13. Optimisation Problems

§13.1. Minimum and maximum

For $R : A \rightarrow A$, $\min R \triangleq \exists n (\epsilon \setminus R^\circ) : EA \rightarrow A$, $\max R \triangleq \min R^\circ$.

$xs \ (\min R) \ x \iff x \in xs \wedge (\forall y \in xs. x R y)$

$xs \ (\min R) \ x \iff (x \text{ in } xs) \text{ and all } x R \setminus : xs \text{ in } q$

$\min R = \exists n \text{ all } R^\circ \text{ all } R \in \setminus R, q \text{'s all; the chains below keep it written } \epsilon \setminus$

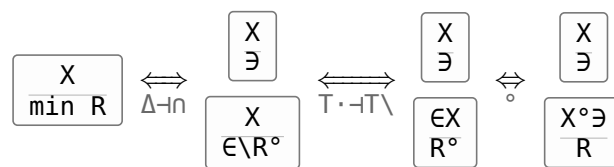
(13.1a)

the law	what it says
$X \subseteq \min R \iff X \subseteq \exists$ and $X^\circ \subseteq \exists R$	in the set, and below every element of it
$\frac{1}{\exists} (\epsilon \setminus R) = R$	bounding a singleton is bounding its element
$\frac{S}{\exists} (\epsilon \setminus R) = S^\circ \setminus R$	bound S 's image without building the set
$\text{union} \triangleq \frac{\exists \exists}{\exists} : E(EA) \rightarrow EA$	flattens a set of sets
$\text{union } (\epsilon \setminus R) = \epsilon \setminus (\epsilon \setminus R)$	bound a union by bounding each member set
$\frac{1}{\exists} \min R = 1 \cap R$	a singleton's minimum is its element, where R is reflexive $\frac{S}{\exists} \min R \text{ at } S := 1$
$\frac{S}{\exists} \min R = S n (S^\circ \setminus R^\circ)$	an S -value that points to every S -value
$\frac{S}{\exists} \min R = \frac{S}{\exists} \min (R n S^\circ S)$	only R between values S gives one argument counts — context
$E(S) \min R = (\exists S) n ((\exists S)^\circ \setminus R^\circ)$	the same for the image of a set $\frac{S}{\exists} \min R \text{ at } S := \exists S$
$P(f) \min R = \min (f R f^\circ) \ f$	shunt a function through a minimum
$P(S) \min R = (\exists S) n (\epsilon \setminus (S R^\circ))$ $R \text{ reflexive}$	fusion with the power relator $\exists$ is the only proof here that tabulates
$P(S) \min R \subseteq (\exists S) n (\epsilon \setminus (S R^\circ))$	the half of the row above that costs nothing
$P(\min R) \min R \subseteq \text{union } \min R$ $R \text{ a preorder}$	a minimum in each set, then a minimum of those
$P(\min R) \min R = P(\text{Dom } (\min R)) \text{ union } \min R$ $R \text{ a preorder}$	the same as an equality, once empty sets are dropped

(13.1b)

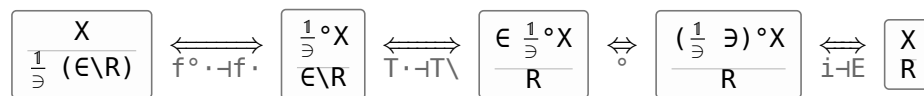
§13.1.1. $X \subseteq \min R \iff X \subseteq \exists$ and $X^\circ \subseteq \exists R$

(13.1.1a)



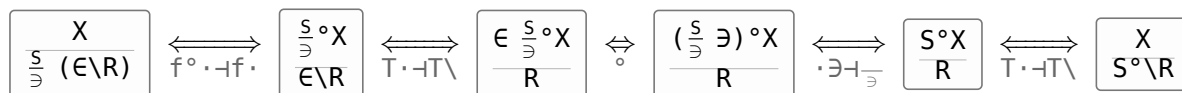
§13.1.2. $\frac{1}{\exists} (\epsilon \setminus R) = R$

(13.1.2a)



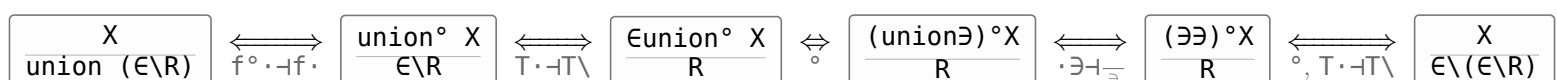
§13.1.3. $\frac{S}{\exists} (\epsilon \setminus R) = S^\circ \setminus R$

(13.1.3a)



§13.1.4. $\text{union } (\epsilon \setminus R) = \epsilon \setminus (\epsilon \setminus R)$

(13.1.4a)



§13.1.5. $\frac{S}{\exists} \min R = S \cap (S^\circ \setminus R^\circ)$

(13.1.5a)

$$\begin{array}{c}
 \boxed{\frac{X}{\frac{S}{\exists} \min R}} \xleftrightarrow{f^\circ \cdot \neg f \cdot} \boxed{\frac{\frac{S}{\exists} X}{\min R}} \xleftrightarrow{\Delta \neg \cap} \begin{array}{c} \boxed{\frac{\frac{S}{\exists} X}{\exists}} \\ \boxed{\frac{\frac{S}{\exists} X}{\in R^\circ}} \end{array} \xleftrightarrow{f^\circ \cdot \neg f \cdot} \begin{array}{c} \boxed{\frac{X}{\frac{S}{\exists} \exists}} \\ \boxed{\frac{X}{\frac{S}{\exists} (\in R^\circ)}} \end{array} \xleftrightarrow{\cdot \exists \neg \frac{S}{\exists}, (13.1.3a)} \begin{array}{c} \boxed{\frac{X}{S}} \\ \boxed{\frac{X}{S^\circ \setminus R^\circ}} \end{array} \xleftrightarrow{\Delta \neg \cap} \boxed{\frac{X}{S \cap (S^\circ \setminus R^\circ)}}
 \end{array}$$

§13.1.6. $\frac{S}{\exists} \min R = \frac{S}{\exists} \min (R \cap S^\circ S)$

(13.1.6a)

$$\begin{array}{c}
 \boxed{\frac{X}{\frac{S}{\exists} \min (R \cap S^\circ S)}} \xleftrightarrow{(13.1.5a), \circ} \boxed{\frac{X}{S \cap (S^\circ \setminus (R^\circ \cap S^\circ S))}} \xleftrightarrow{\Delta \neg \cap, T \cdot \neg T \setminus} \begin{array}{c} \boxed{\frac{X}{S}} \\ \boxed{\frac{S^\circ X}{R^\circ \cap S^\circ S}} \end{array} \\
 \\
 \xleftrightarrow{\Delta \neg \cap, S^\circ \cdot \text{monotone}} \begin{array}{c} \boxed{\frac{X}{S}} \\ \boxed{\frac{S^\circ X}{R^\circ}} \end{array} \xleftrightarrow{T \cdot \neg T \setminus, \Delta \neg \cap} \boxed{\frac{X}{S \cap (S^\circ \setminus R^\circ)}} \xleftrightarrow{(13.1.5a)} \boxed{\frac{X}{\frac{S}{\exists} \min R}}
 \end{array}$$

§13.1.7. $P(f) \min R = \min(f R f^\circ) f$

(13.1.7a)

$$\begin{array}{c}
 \boxed{\frac{P(f) \min R}{f \text{ a map}}} \xrightarrow[\text{P=E on maps, (13.1.5a)}]{=} \boxed{(\exists f) \cap ((\exists f)^\circ \setminus R^\circ)} \xrightarrow[\circ, T \cdot \neg T \setminus, f^\circ \cdot \neg f \cdot]{=} \boxed{(\exists f) \cap (\in \setminus (f R^\circ))} \\
 \\
 \xrightarrow[\text{modular law, f simple}]{=} \boxed{(\exists \cap ((\in \setminus (f R^\circ)) f^\circ)) f} \xrightarrow[\cdot f \neg \cdot f^\circ, \circ, \min]{=} \boxed{\min(f R f^\circ) f}
 \end{array}$$

§13.1.8. $P(S) \min R \in (\exists S) \cap (\in \setminus (S R^\circ))$

(13.1.8a)

$$\begin{array}{c}
 \boxed{\frac{P(S) \min R}{(\exists S) \cap (\in \setminus (S R^\circ))}} \xleftrightarrow{\Delta \neg \cap, T \cdot \neg T \setminus} \begin{array}{c} \boxed{\frac{P(S) \min R}{\exists S}} \\ \boxed{\frac{\in P(S) \min R}{S R^\circ}} \end{array} \xleftrightarrow{\text{UP of min}} \begin{array}{c} \boxed{\frac{P(S) \exists}{\exists S}} \\ \boxed{\frac{\in P(S)}{S \in}} \end{array}
 \end{array}$$

§13.1.9. $P(\min R) \min R \in \text{union} \min R$

(13.1.9a)

$$\begin{array}{c}
 \boxed{\frac{P(\min R) \min R}{\text{union} \min R}} \xleftrightarrow{(13.1.5a)} \boxed{\frac{P(\min R) \min R}{(\exists \exists) \cap ((\exists \exists)^\circ \setminus R^\circ)}} \xleftrightarrow{\circ, \Delta \neg \cap, T \cdot \neg T \setminus} \begin{array}{c} \boxed{\frac{P(\min R) \min R}{\exists \exists}} \\ \boxed{\frac{\in \in P(\min R) \min R}{R^\circ}} \end{array} \xleftrightarrow{\text{UP of min, R transitive}} \begin{array}{c} \boxed{\frac{P(\min R) \exists}{\exists \min R}} \\ \boxed{\frac{\in P(\min R)}{\min R \in}} \end{array}
 \end{array}$$

§13.2. Lax natural transformations (LaT)

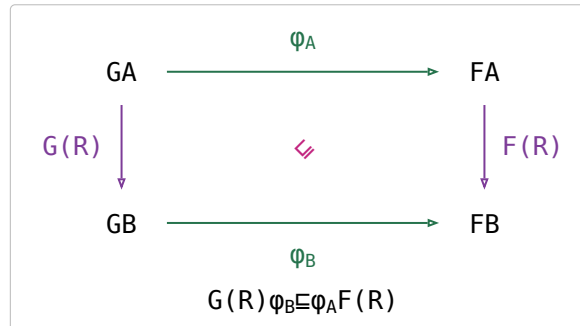
For relators $G, F : \mathcal{C} \rightarrow \mathcal{D}$ and components $\varphi_A : GA \rightarrow FA$, φ is **lax at** $R : A \rightarrow B$ when $G(R)\varphi_B \sqsubseteq \varphi_A F(R)$ — both sides $GA \rightarrow FB$, the left through the component at B , the right through the one at A .

(13.2a)

$\varphi : G \Rightarrow F$ is a **lax natural transformation** (LaT) when it is lax at **every** R . (5.13)

$\exists : P \Rightarrow \text{Id}$ — that is $P(R)\exists \sqsubseteq \exists R$, (11.3d)'s first row at $X := P(R)$.

Lax at every **map** already gives LaT, and at a map the inequation is an equality $G(f)\varphi = \varphi F(f)$: laxness is about relations only. Theorem 5.2



(13.2b)

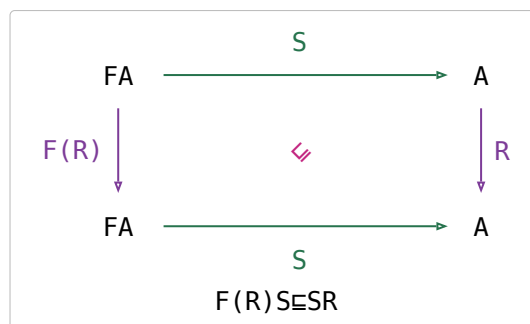
§13.3. Monotonic algebras

An F-algebra $S : FA \rightarrow A$ is **monotonic on** $R : A \rightarrow A$ when it is lax at R at $G := F$, $F := \text{Id}$, $\varphi := S$: $F(R)S \sqsubseteq SR$. R is an **endorelation**, so $B = A$ and the two components φ_A, φ_B are the one arrow S — an algebra, not a family.

(13.3a)

For a map $f : FA \rightarrow A$ that is $f \circ F(R)f \sqsubseteq R$, equivalently $F(R) \sqsubseteq fRf^\circ$.

$(\leq \times \leq) + \sqsubseteq + \leq$ — addition on Nat is monotonic on $\leq$, which at the point level reads $c = a + b \wedge a \leq a' \wedge b \leq b' \implies c \leq a' + b'$.

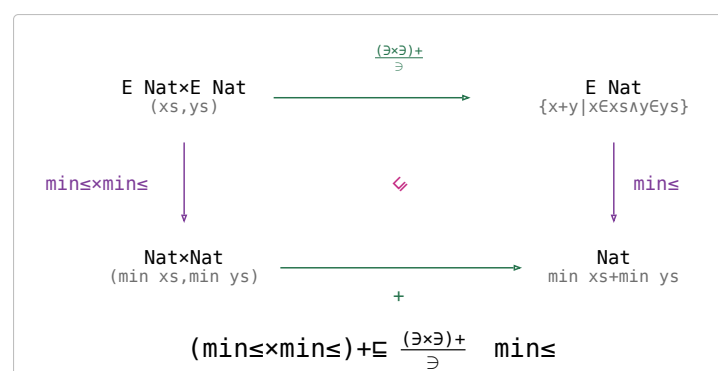
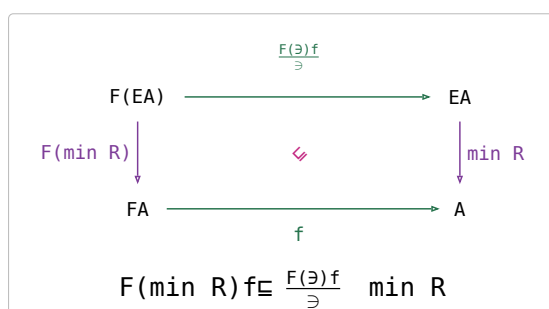


(13.3b)

$f : FA \rightarrow A$ **distributes over** R if $F(\min R)f \sqsubseteq \frac{F(\exists)f}{\exists} \min R$.

(13.3c)

$+$ distributes over $\leq$, at the point level $\min xs + \min ys = \min\{x+y \mid x \in xs \wedge y \in ys\}$ for xs, ys non-empty and $\min \triangleq \min \leq$.



(13.3d)

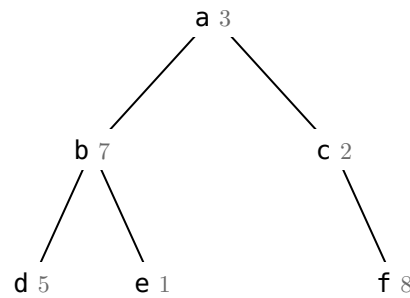
the law	what it says	(13.3e)
$f \circ F(R) f \in R \iff f \circ F(R^\circ) f \in R^\circ$	a map is monotonic on an order and on its opposite together — a relation is not	
$f \circ F(R) f \in R \iff F(\min R) f \in \frac{F(\exists) f}{\exists} \min R$ Theorem 7.1, f a map	for a map the two definitions above are one condition	
$(\frac{S}{\exists} \min R) \in \frac{(\exists)}{\exists} \min R$ Theorem 7.2, R a preorder and $F(R^\circ) S \in SR^\circ$	the greedy theorem : keeping one minimum at every step beats no more than taking the minimum at the end	

§13.4. Planning a company party

definition	type	note	(13.4a)
$\text{tree } A ::= \text{node } (A, [\text{tree } A])$	$\text{tree} : A \rightarrow A$	The company hierarchy: an employee, and the list of subtrees under them.	
$F(A, B) = A \times [B]$	$F : A \times A \rightarrow A$	The base functor tree folds: an employee beside the recursive position, one layer deep.	
rating	rating : $A \rightarrow \text{Real}$	What one employee is worth as a guest.	
$\text{cost} \triangleq \text{list}(\text{rating}) \text{ sum}$	cost : $[A] \rightarrow \text{Real}$	What a guest list is worth.	
$R \triangleq \text{cost} \leq \text{cost}^\circ$	$R : [A] \rightarrow [A]$	The preorder the guest list is maximised over.	
$\text{choose} \triangleq \pi_1 \cup \pi_2$	choose : $[A] \times [A] \rightarrow [A]$	Takes one of the two parties a subtree returns.	
$\text{include} \triangleq (1 \times (\text{list}(\pi_2) \text{ concat})) \text{ cons}$	include : $F(A, [A] \times [A]) \rightarrow [A]$	The party that invites the root, which puts every immediate subtree's root out. A map.	
$\text{exclude} \triangleq (1 \times (\text{list}(\text{choose}) \text{ concat})) \pi_2$	exclude : $F(A, [A] \times [A]) \rightarrow [A]$	The party that leaves the root out, so each subtree is free to choose. Not a map.	
$S \triangleq (\text{include}, \text{exclude})$	$S : F(A, [A] \times [A]) \rightarrow [A] \times [A]$	The algebra: one step returns both parties of a subtree at once.	
$\text{party} \triangleq (\exists) \text{ choose}$	party : $\text{tree } A \rightarrow [A]$	Every guest list the president's ruling allows.	
the specification $\frac{\text{party}}{\exists} \max R$	$\frac{\text{party}}{\exists} \max R : \text{tree } A \rightarrow [A]$	A guest list of greatest total conviviality.	

§13.4.1. include and exclude

(13.4.1a)



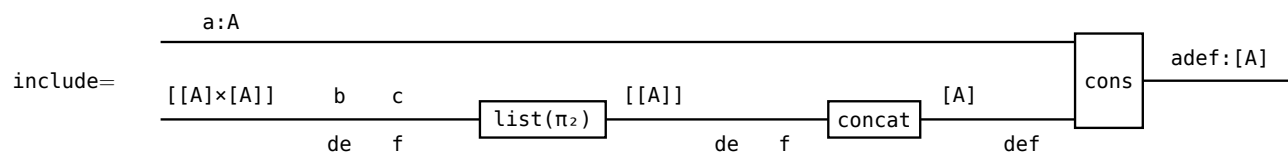
the small grey number is the employee's rating

(13.4.1b)

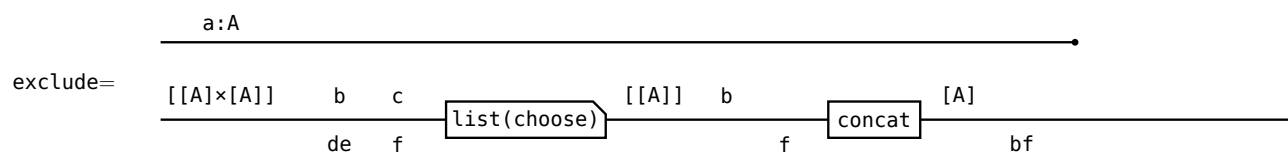
node	the algebra's input $A \times [[A] \times [A]]$	include	exclude
d, e, f	$(d, [])$	$[d]$	$[]$
b	$(b, ([([d], []), ([e], [])]))$	$[b]$	$[d, e], [d], [e], []$
c	$(c, ([([f], [])]))$	$[c]$	$[f], []$
a	$(a, ([([b], ...), ([c], ...)]))$	$[a] \# \text{exclude}(b) \# \text{exclude}(c)$	$\text{choose}(b) \# \text{choose}(c)$

with those ratings, $\max R$ keeps $([b]=7, [d, e]=6)$ at b , where **include** wins, and $([c]=2, [f]=8)$ at c , where **exclude** wins, so at the root **include** $= [a, d, e, f] = 17$ beats **exclude** $= [b, f] = 15$, and **choose** takes 17.

(13.4.1c)



(13.4.1d)



list(choose) $[[([d], []), ([e], [])]$
 1st item $[[d], []]$ choose $\rightarrow [d]$ or $[]$
 2nd item $[[e], []]$ choose $\rightarrow [e]$ or $[]$
 $2 \times 2 = 4$ combinations: $[[d], [e]]$ $[[d], []]$ $[[], [e]]$ $[[], []]$
 concat flattens: $[d, e]$ $[d]$ $[e]$ $[]$

(13.4.1e)

§13.4.2. list($R \times R$)

$[([b], [d]) , ([c], [])]$
 $\downarrow$
 $[([b], [d, e]) , ([c], [f])]$ list($R \times R$), elementwise

(13.4.2a)

$([b], [d])$
 $\begin{cases} \text{exclude: the best party in b's subtree when b does not come} \\ \text{include: the best party in b's subtree when b comes} \end{cases}$

(13.4.2b)

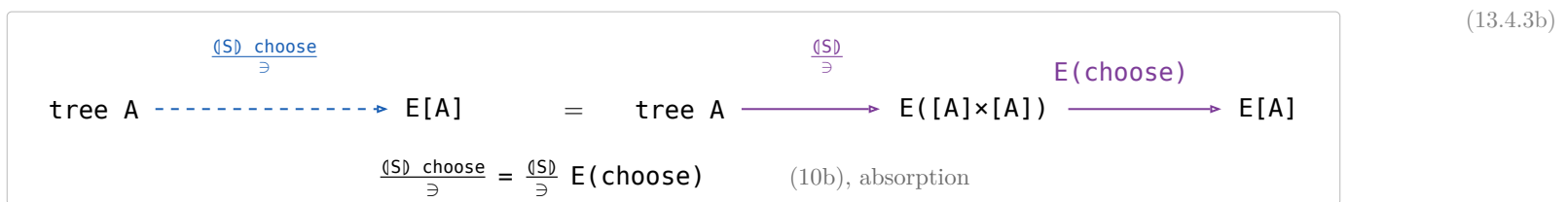
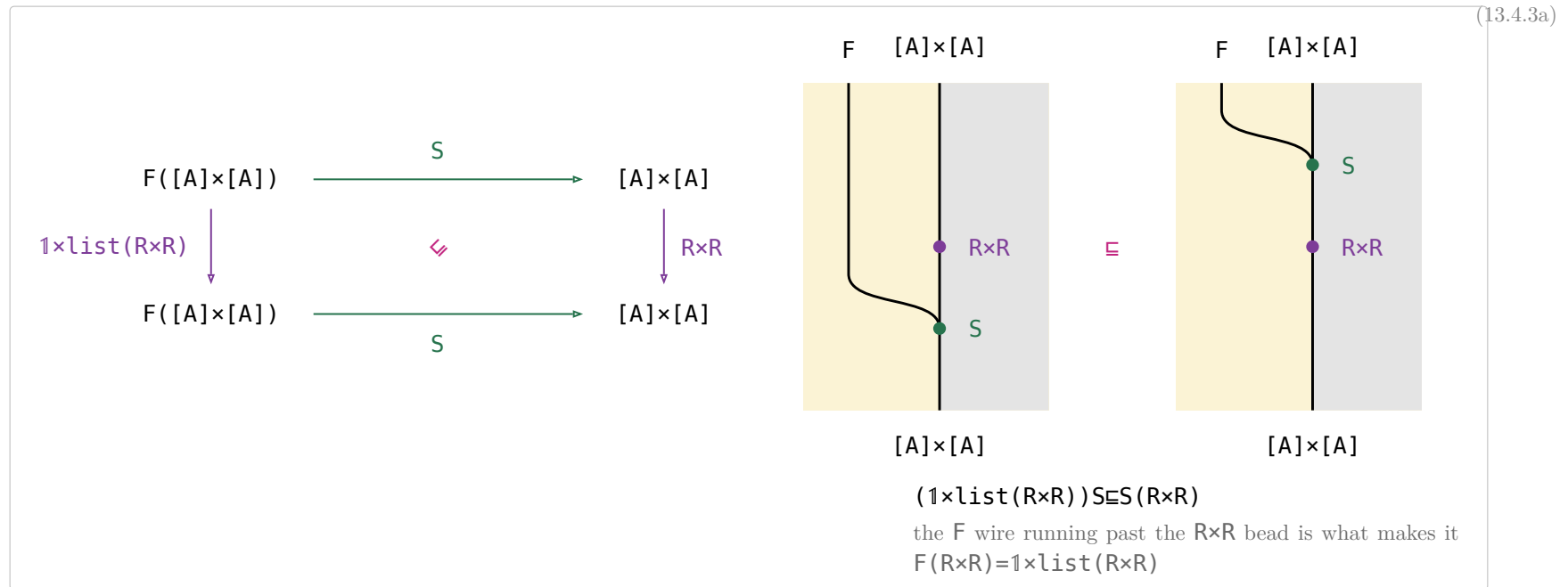
element	1st component include	2nd component exclude
1st, $([b], [d])$	$[b] R [b]$ $7 \leq 7$, reflexivity	$[d] R [d, e]$ $5 \leq 6$, and $([b], [d, e])$ is the pair $\max R$ keeps at b
2nd, $([c], [])$	$[c] R [c]$ $2 \leq 2$, reflexivity	$[] R [f]$ $0 \leq 8$, and $([c], [f])$ is the pair $\max R$ keeps at c

the first component of $1 \times \text{list}(R \times R)$ is the root employee, a 1 a : an employee carries no cost order.

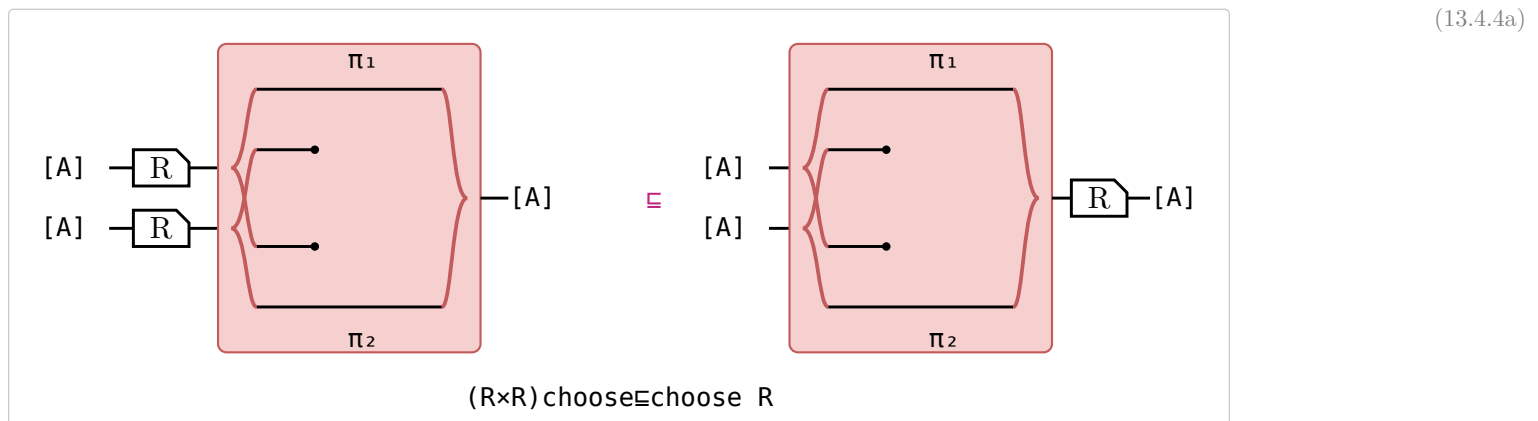
$R \triangleq \text{cost} \leq \text{cost}^\circ$ is a preorder, so the components that do not move are related by reflexivity — $R \times R$ demands a relation in **both** components, it cannot skip one.

R compares cost only, not contents: $[] \ R \ [f]$ is legal though $[f]$ has an element $[]$ does not. That is why the three leaves of §13.4.3's proof all reduce to “**cost** is a sum”.

§13.4.3. $(1 \times \text{list}(R \times R)) S \sqsubseteq S(R \times R) \text{ — } S : F([A] \times [A]) \rightarrow [A] \times [A] \text{ monotonic on } R \times R$



§13.4.4. $(R \times R) \text{ choose } \sqsubseteq \text{choose } R \text{ — } \text{choose monotonic on } R$



(13.4.4b)

$$\boxed{(R \times R) \text{ choose}} \xrightarrow{\text{choose} \triangleq \pi_1 \cup \pi_2} \boxed{(R \times R) (\pi_1 \cup \pi_2)} \xrightarrow{\text{composition distributes over } \cup} \boxed{(R \times R) \pi_1 \cup (R \times R) \pi_2}$$

$$\xrightarrow{(5.2) \text{ of } (11.1.2b), R \times S = \langle \pi_1 R, \pi_2 S \rangle} \boxed{\langle \pi_1 R, \pi_2 R \rangle \pi_1 \cup \langle \pi_1 R, \pi_2 R \rangle \pi_2}$$

$$\xrightarrow{(5.6) \text{ and } (5.7) \text{ of } (11.1.2b)} \boxed{(\text{Dom } (\pi_2 R)) \pi_1 \cup (\text{Dom } (\pi_1 R)) \pi_2 R}$$

$$\text{Dom } \sqsubseteq 1 \text{ — the only step that loses, and the whole law } \boxed{\pi_1 R \cup \pi_2 R}$$

$$\xrightarrow{\text{composition distributes over } \cup} \boxed{(\pi_1 \cup \pi_2) R} \xrightarrow{\text{choose} \triangleq \pi_1 \cup \pi_2} \boxed{\text{choose } R}$$

§13.4.5. The derivation

(13.4.5a)

$$\boxed{\text{party} \supseteq \max R} \xrightarrow{\text{definition of party}} \boxed{\frac{(S) \text{ choose}}{\supseteq} \max R} \xrightarrow{(10b), \text{ absorption}} \boxed{\frac{(S)}{\supseteq} E(\text{choose}) \max R}$$

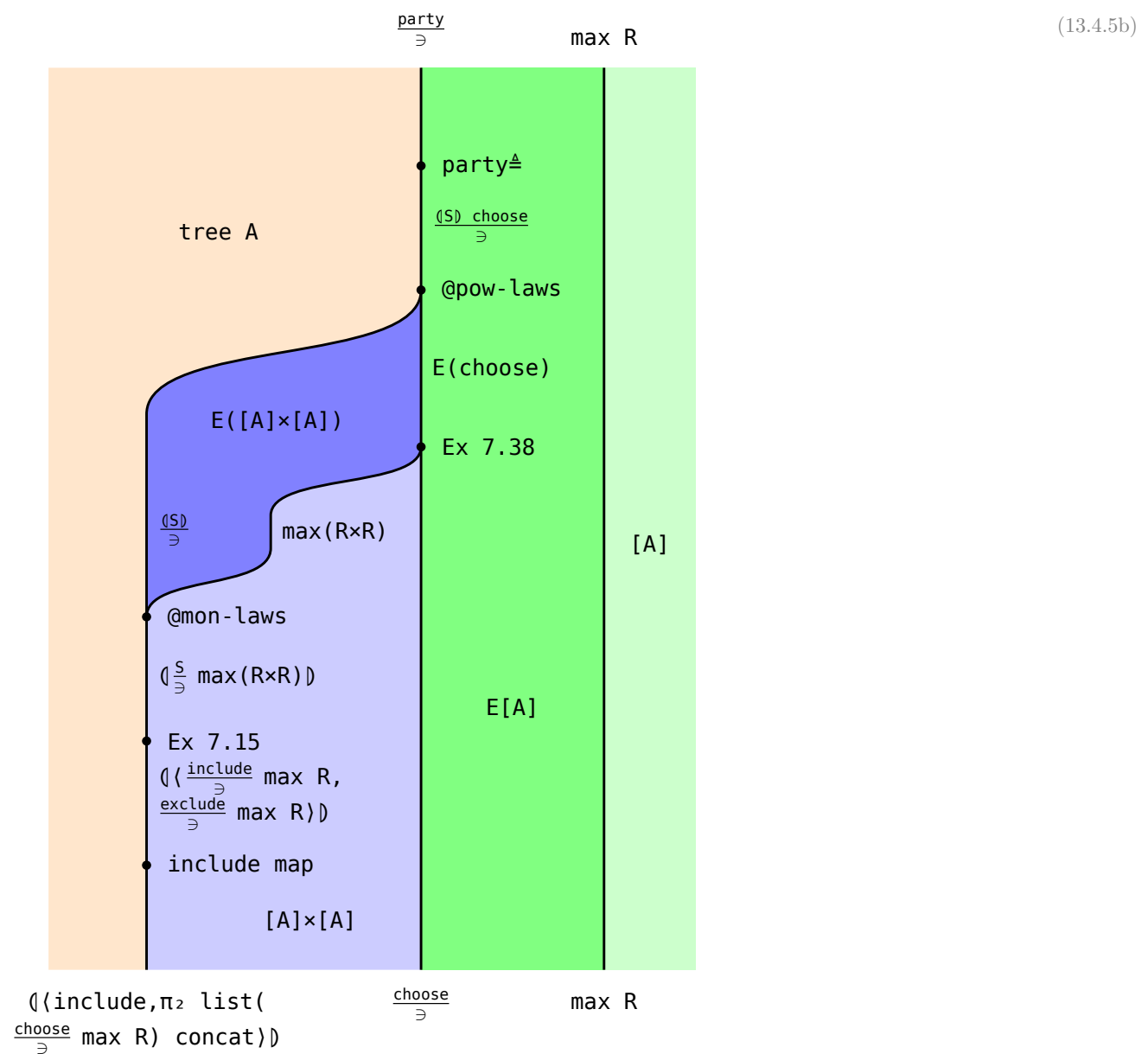
$$\text{Ex 7.38, §13.4.4} \quad \boxed{\frac{\text{Ex 7.38}}{\text{Ex 7.38}} \max(R \times R) \xrightarrow{\text{choose}} \max R}$$

$$(13.3e), (1 \times \text{list}(R \times R)) S \subseteq S(R \times R) \quad \boxed{\frac{\text{Ex 7.38}}{\text{Ex 7.38}} \max(R \times R) \xrightarrow{\text{choose}} \max R}$$

$$\text{Ex 7.15, the fork splits} \quad \boxed{\frac{\text{Ex 7.15}}{\text{Ex 7.15}} \left(\frac{\text{include}}{\text{include}} \max R, \frac{\text{exclude}}{\text{exclude}} \max R \right) \xrightarrow{\text{choose}} \max R}$$

$$\text{include a map, } \max R \text{ into each branch} \quad \boxed{\frac{\text{include a map, } \max R \text{ into each branch}}{\text{include a map, } \max R \text{ into each branch}} \left(\frac{\text{include}}{\text{include}} \max R, \frac{\text{exclude}}{\text{exclude}} \max R \right) \xrightarrow{\text{choose}} \max R}$$

the program is $\text{party} = \frac{\text{choose}}{\text{choose}} \max R$ with **exclude** renamed to $\pi_2 \text{ list}(\frac{\text{choose}}{\text{choose}} \max R) \text{ concat}$ — **include** was already a map, and **exclude** becomes one.



include map	⊆ include a map, $\max R$ into each branch
Ex 7.15	⊆ the fork splits
@mon-laws	⊆ $(1 \times \text{list}(R \times R)) S \subseteq S(R \times R)$
Ex 7.38	⊆ $(R \times R) \text{ choose} \subseteq \text{choose } R$
@pow-laws	= absorption
party ^Δ	= definition of party

Marsden's conventions, not the note's: composition runs left to right and 2-cells run upward, so the picture is the derivation upside down — the bottom edge is the program, the top edge the specification, and each bead says that what is below it is contained in what is above.

§13.5. Shortest paths on a cylinder

definition	type	note	(13.5a)
$F(A, X) = A + A \times X$	$F : \mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$	The base functor: one square of the new column, alone or in front of the path so far.	
$L = \text{list}^+$	$L : \mathcal{A} \rightarrow \mathcal{A}$	A path is a non-empty list of squares, one per column crossed.	
α the initial algebra	$\alpha : F(A, LA) \rightarrow LA$	$\alpha = [\text{wrap}, \text{cons}]$: a path is started by one square, or extended by one.	
N the n-tuple relator	$N : \mathcal{A} \rightarrow \mathcal{A}$	One component per row of a column, so the fold carries n answers at once.	
$R \triangleq \text{sum} \leq \text{sum}^\circ$	$R : L \text{ Nat} \rightarrow L \text{ Nat}$	The cost of a path, which the cheapest minimises.	
setify	setify : $NA \rightarrow EA$	Forgets which row a component came from, so the n answers become one set.	
moves	moves : $NA \rightarrow E(NA)$	The three columns a path may step to: rotated up, unrotated, rotated down.	
trans	trans : $E(NA) \rightarrow N(EA)$	Transposes a set of tuples, so every row collects its own paths.	
zip	zip : $F(NA, NB) \rightarrow NF(A, B)$	Commutates N with the base functor, pairing each new square with its own row's paths.	
$cp \triangleq \frac{F(1, \exists)}{\exists}$	cp : $F(A, EB) \rightarrow E(F(A, B))$	Turns a square paired with a set of paths into the set of extensions of those paths.	
generate $\triangleq F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp } P(\alpha))$	generate : $F(NA, N(E(LA))) \rightarrow N(E(LA))$	One fold step: extends every path of every row by the new column.	
paths $\triangleq (\text{generate}) \text{ setify union}$	paths : $L \text{ N Nat} \rightarrow E(L \text{ Nat})$	Every path across the cylinder.	
the specification paths min R	paths min R : $L \text{ N Nat} \rightarrow L \text{ Nat}$	A cheapest path from the entry side to the exit side.	

§13.5.1. $N(E(L \ A))$

```

1  5
2  6
3  7      [(1,2,3,4), (5,6,7,8)] : L N Nat
4  8

```

(13.5.1a)

```

(generate)[(5,6,7,8)] = ({[5]}, {[6]}, {[7]}, {[8]})

```

(13.5.1b)

```

generate((1,2,3,4), ({[5]}, {[6]}, {[7]}, {[8]})) =
( {[1,5], [1,6], [1,8]},
  {[2,5], [2,6], [2,7]},
  {[3,6], [3,7], [3,8]},
  {[4,5], [4,7], [4,8]} )

```

N	the 4-tuple, one component per row of the entry column	(13.5.1c)
E	the set of paths that can start in that row	
L	a path is the non-empty list of the squares it crosses	

row 1's set has no $[1, 7]$ — from row 1 only rows 4, 1 and 2 are reachable — and it does have $[1, 8]$, because the cylinder glues the bottom row to the top. That is $\text{moves trans } N(\text{union})$ of (13.5a): component k collects the paths of rows $k-1, k, k+1$.

§13.5.2. The derivation

$$\begin{array}{c}
 \boxed{\text{paths min } R} \xrightarrow{\text{definition of paths}} \boxed{\langle \text{generate} \rangle \text{ setify union min } R} \xrightarrow{(13.1b) \text{ at } \overline{\text{union}}, R \text{ a preorder}} \boxed{\langle \text{generate} \rangle \text{ setify } P(\text{min } R) \text{ min } R} \quad (13.5.2a) \\
 \\
 \text{setify } \overline{\text{lax natural}} \xrightarrow{\overline{\text{setify}}} \boxed{\langle \text{generate} \rangle N(\text{min } R) \text{ setify min } R} \xrightarrow{(11.4.2a) \text{ at } Q} \boxed{\langle Q \rangle \text{ setify min } R} \\
 \\
 \boxed{\text{generate } N(\text{min } R)} \xrightarrow{(7.13), \text{ then zip, trans, moves lax natural}} \boxed{F(1, N(\text{min } R)) Q} \\
 \\
 \boxed{Q} \xrightarrow{\text{the fusion condition read as a definition}} \boxed{F(1, \text{moves trans } N(\text{min } R)) \text{ zip } N(\alpha)} \\
 \\
 \xrightarrow{\text{zip}=\mathbb{1}+\text{zip}', \alpha=[\text{wrap}, \text{cons}]} \boxed{[N(\text{wrap}), (1 \times \text{moves trans } N(\text{min } R)) \text{ zip}' N(\text{cons})]}
 \end{array}$$

(7.13) is $F(1, \text{min } R) \alpha \sqsubseteq_{\text{cp}} P(\alpha) \text{ min } R$, Theorem 7.1 at the map α with $\frac{F(1, \exists) \alpha}{\exists} = \text{cp } P(\alpha)$: extending every path in a set and then taking a minimum is beaten by extending one minimum. It is the crux here, not the greedy theorem.

§13.6. The security van problem

(13.6a)

definition	type	note
$R \triangleq \text{length} \leq \text{length}^\circ$	$R : [[\text{Int}]] \rightarrow [[\text{Int}]]$	The order the schedule is minimised over: fewer van visits is better.
$\text{ceiling} \triangleq \frac{\text{prefix sum}}{\ni} \max \leq$	$\text{ceiling} : [\text{Int}] \rightarrow \text{Int}$	The highest the bank's balance reaches over a stretch of transactions.
$\text{floor} \triangleq \frac{\text{prefix sum}}{\ni} \min \leq$	$\text{floor} : [\text{Int}] \rightarrow \text{Int}$	The lowest it reaches, so $\text{ceiling} - \text{floor}$ is the cash the stretch has to carry.
secure the coreflexive on x with $\text{bmax}(\text{ceiling } x, \text{ceiling } x - \text{floor } x) \leq N$	$\text{secure} : [\text{Int}] \rightarrow [\text{Int}]$	The stretches one van visit can serve: some starting reserve keeps the cash between 0 and N.
ok the coreflexive on (a, xs) with xs non-empty and $[a] \# \text{head } xs$ secure	$\text{ok} : \text{Int} \times [[\text{Int}]] \rightarrow \text{Int} \times [[\text{Int}]]$	The test the final program runs, in place of old 's secure .
$\text{new} \triangleq (\text{wrap} \times 1) \text{ cons}$	$\text{new} : \text{Int} \times [[\text{Int}]] \rightarrow [[\text{Int}]]$	Calls the van: the transaction opens a segment of its own.
$\text{glue} \triangleq (1 \times \text{cons}^\circ) \text{ assocl } (\text{cons} \times 1) \text{ cons}$	$\text{glue} : \text{Int} \times [[\text{Int}]] \rightarrow [[\text{Int}]]$	Puts the transaction on the front of the first segment.
$\text{old} \triangleq (1 \times \text{cons}^\circ) \text{ assocl } ((\text{cons } \text{secure}) \times 1) \text{ cons}$	$\text{old} : \text{Int} \times [[\text{Int}]] \rightarrow [[\text{Int}]]$	glue restricted to a first segment that stays secure.
$\text{partition} = ([\text{nil}, \text{newu glue}])$	$\text{partition} : [\text{Int}] \rightarrow [[\text{Int}]]$	Every splitting of the transactions into consecutive segments.
$S \triangleq [\text{nil}, \text{newu old}]$	$S : 1 + \text{Int} \times [[\text{Int}]] \rightarrow [[\text{Int}]]$	The algebra whose fold is every splitting of the transactions into secure segments.
$\text{partition list}(\text{secure}) = (S)$	$\text{partition list}(\text{secure}) : [\text{Int}] \rightarrow [[\text{Int}]]$	Fusion: keeping only secure segments is what turns glue into old .
$H \triangleq (\text{head prefix}^\circ \text{ head}^\circ) u (\text{nil}^\circ \text{ nil})$	$H : [[\text{Int}]] \rightarrow [[\text{Int}]]$	One schedule's first segment is a prefix of the other's, or both are empty.
$R; H \triangleq R n (R^\circ \Rightarrow H)$	$R; H : [[\text{Int}]] \rightarrow [[\text{Int}]]$	The refinement of R that makes both halves of S monotonic.
$ R \triangleq R n \neg R^\circ$	$ R : [[\text{Int}]] \rightarrow [[\text{Int}]]$	The strict part R splits into: $R; H = R u (R n H)$.
the specification $\frac{\text{partition list}(\text{secure})}{\ni} \min R$	$\frac{\text{partition list}(\text{secure})}{\ni} \min R : [\text{Int}] \rightarrow [[\text{Int}]]$	A schedule with the fewest secure segments.

(13.6b)

$$\boxed{\frac{\text{partition list}(\text{secure})}{\ni} \min R} \quad (11.4.2a), \quad \text{secure} \quad \text{prefix} \sqsubseteq \text{prefix} \quad \boxed{\frac{(S)}{\ni} \min R} \quad (7.14) \text{ holds, } (7.15) \text{ FALSE; } R; H \in R \quad \boxed{\frac{(S)}{\ni} \min(R; H)}$$

(13.3e), S monotonic on $\overline{\mathbf{R}};H$ by (7.16) and (7.17) $\boxed{(\frac{S}{\supset} \min(\mathbf{R};H))\Downarrow}$ $\text{old} \sqsubseteq \overline{\text{new}} (\mathbf{R};H) \circ \boxed{(\llbracket \text{nil}, (\text{ok} \rightarrow \text{glue}, \text{new}) \rrbracket)\Downarrow}$

$\boxed{(\mathbb{1} \times (\mathbf{R};H)) \text{ old}}$ $\overline{\overline{\mathbf{R};H = |\mathbf{R}| \cup (\mathbf{R} \cap H), \text{ } \cup \text{ distributes}}}$ $\boxed{(\mathbb{1} \times |\mathbf{R}|) \text{ old} \cup (\mathbb{1} \times (\mathbf{R} \cap H)) \text{ old}}$ $(7.19) \text{ and } (7.20) \text{ on } |\mathbf{R}|, (7.21) \text{ on } \mathbf{R} \cap H \sqsubseteq \boxed{\text{new } (\mathbf{R} \cap H) \text{ old } (\mathbf{R} \cap H)}$

$X \cap Y \sqsubseteq X;Y, \text{ converses } \sqsubseteq \boxed{(\text{new} \cup \text{old})(\mathbf{R};H)}$

(7.15) $(\mathbb{1} \times \mathbf{R}) \text{ old} \sqsubseteq (\text{new} \cup \text{old})\mathbf{R}$ is FALSE — the shorter partition need not stay secure — and that is what refining $\mathbf{R}$ to $\mathbf{R};H$ buys; the second chain is (7.17), and (7.16) rests the same way on (7.18) $(\mathbb{1} \times \mathbf{T}) \text{ new} \sqsubseteq \text{new } H$.

§14. Thinning Algorithms

§14.1. Thinning

For $Q : A \rightarrow A$, $\text{thin } Q \triangleq (\exists/\exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : EA \rightarrow EA$ (8.1).

$ys \text{ (thin } Q) \text{ } xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \text{ } Q \text{ } a)$

(14.1a)

the law	what it says
$X \subseteq \frac{S}{\exists} \text{ thin } Q \iff X \subseteq S \text{ and } S^\circ X \subseteq Q^\circ \epsilon$	the universal property: everything kept is an S -value, and every S -value has a Q -lower bound among the kept ones
$Q \subseteq R \implies \text{thin } Q \subseteq \text{thin } R$	the fewer pairs Q relates, the fewer subsets count as thinnings
$1 \in \text{thin } Q$, and $\text{thin } Q$ is a preorder if Q is	keeping everything is always a legal thinning
$\min R = \text{thin } Q \min R$ $Q \subseteq R$, both preorders	thin-introduction : thinning first cannot lose an R -minimum
$\text{thin } Q \supseteq \min Q \frac{1}{\exists}$ (8.2)	thin-elimination : keeping one element is a thinning, but its domain is the sets $\min Q$ is defined on
$\frac{S}{\exists} \text{ thin } Q \supseteq \frac{S}{\exists} \min R \frac{1}{\exists}$ (8.3), $R \cap (S^\circ S) \subseteq Q$	the usable variant: R need only refine Q between values S gives one argument
$\text{union thin } Q \supseteq P(\text{thin } Q) \text{ union}$ (8.4)	thinning each member set is a thinning of the union
$Q(\frac{F(\exists,1)S}{\exists} \text{ thin } Q) \subseteq \frac{(S)}{\exists} \text{ thin } Q$ Theorem 8.1, S monotonic on Q	the thinning theorem : thinning at every step beats thinning only at the end
$Q(\frac{F(\exists,1)S}{\exists} \text{ thin } Q) \min R \subseteq \frac{(S)}{\exists} \min R$ Corollary 8.1, $Q \subseteq R$ as well	the same against the optimisation problem itself, by thin-introduction

(14.1b)

§14.2. Paths in a layered network

$F(A, X) = A + A \times X$, $L = \text{list}^+$ with initial algebra $\alpha \triangleq [\text{wrap}, \text{cons}] : F(A, LA) \rightarrow LA$.

$\text{wrapz} \triangleq (\text{wrap}, \text{zero})$, $\text{consw } (a, (xs, n)) = (\text{cons } (a, xs), \text{wt } (a, \text{head } xs) + n)$.

$\text{cost} \triangleq ([\text{wrapz}, \text{consw}]) \pi_2$, $Q([\text{wrapz}, \text{consw}]) = (1, \text{cost})$, $R \triangleq \text{cost} \leq \text{cost}^\circ$.

$Q \triangleq R \cap (\text{head } \text{head}^\circ)$, $S \triangleq F(1, \exists) \alpha$, $\frac{F(\exists, 1)}{\exists} = 1 + \text{cpl}$, $\frac{F(1, \exists)}{\exists} = 1 + \text{cpr}$, $\text{step} \triangleq \text{cpr } P(\text{cons}) \min R$.

(14.2a)

the law	what it says
$\frac{Q(F(\exists, 1) \alpha)}{\exists} \min R = \frac{L(\exists)}{\exists} \min R$	the specification: a least-cost path across the network
$F(\exists, Q) \alpha \subseteq F(\exists, 1) \alpha Q$	S is monotonic on Q ; on R it is not, since the next edge can cost arbitrarily much
$Q(\frac{F(\exists, \exists) \alpha}{\exists} \text{ thin } Q) \min R \subseteq \frac{Q(F(\exists, 1) \alpha)}{\exists} \min R$ Corollary 8.1	thinning applies: one partial path per starting vertex is enough
$S \text{ head} \in [1, \pi_2]$ $S \text{ head}$ simple, so $R \cap (S^\circ S) \subseteq Q$	the side condition of (8.3): between two paths S builds from one argument, equal cost and equal head already means Q
$\frac{F(\exists, \exists) \alpha}{\exists} \text{ thin } Q$ $\supseteq \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)}{\exists} P(\alpha) \min R)$ (8.4), (8.3), $P(\frac{1}{\exists}) \text{ union} = 1$	thin eliminated: split the algebra at $F(\exists, 1)F(1, \exists)$, thin each part, one minimum per part
$\frac{F(1, \exists)}{\exists} P(\alpha) \min R = [\text{wrap}, \text{step}]$	that inner part read off the two branches of α
$\frac{Q(F(\exists, 1) \alpha)}{\exists} \min R \supseteq Q([P(\text{wrap}), \text{cpl } P(\text{step})]) \min R$	the program: one fold, a best path per vertex of the current layer

(14.2b)

§14.3. Implementing thin

(14.3a)

$\text{setify} : [A] \rightarrow EA$, $\text{cup} : EA \times EA \rightarrow EA$, $\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$, $\text{listcp}(F) : F([A]) \rightarrow [FA]$, $\text{sort } P \triangleq \text{setify}^\circ \text{ ordered } P$ for P a connected preorder.

$\text{thinlist } Q$ is any $\text{thinlist } Q \subseteq \text{subseq}$ with $\text{thinlist } Q \text{ setify} \subseteq \text{setify thin } Q$; one is $\langle [\text{nil}, \text{bump } Q] \rangle$, $\text{bump } Q (a, []) = [a]$, $\text{bump } Q (a, [b] \# xs) = (b \ Q \ a \rightarrow [a] \# xs, a \ Q \ b \rightarrow [b] \# xs, [a] \# [b] \# xs)$.

Binary thinning data: $S = (f_1 p_1) \cup (f_2 p_2)$ with p_1, p_2 coreflexive; Q a preorder with $Q \subseteq R$ and both $f_1 p_1, f_2 p_2$ monotonic on Q ; P a connected preorder with both f_1, f_2 monotonic on P ; $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$.

the law	what it says	(14.3b)
$\text{thinlist } Q \text{ xs} = [\text{minlist } Q \text{ xs}]$ (8.5), Q connected, XS non-empty	what thinning should come to when it can: one element	
$\text{sort } P \text{ thinlist } Q \subseteq \text{thin } Q \text{ sort } P$ (8.6)	thinning a sorted list is a thinning of the set — this is what $\text{thinlist } Q \subseteq \text{subseq}$ buys	
$\text{sort } P \text{ minlist } Q \subseteq \text{min } Q$ (8.7)	a minimum of the sorted list is a minimum of the set	
$\text{sort}(f P f^\circ) \text{ list}(f) \subseteq P(f)$ $\text{sort } P$ (8.8)	shunt a function through a sort	
$\text{sort } P \text{ filter}(p) \subseteq E(p) \text{ sort } P$ (8.9), p coreflexive	filtering a sorted list sorts the restricted set	
$(\text{sort } P \times \text{sort } P) \text{ merge } P \subseteq \text{cup sort } P$ (8.10)	merging two sorted lists sorts their union	
$F(\text{sort } P) \text{ listcp}(F) \subseteq \text{cp}(F) \text{ sort}(FP)$ (8.11), F linear	$\text{listcp}(F)$ is the list implementation of the cartesian product $\text{cp}(F)$	
$F(\text{sort } P) \text{ listcp}(F) \text{ list}(f) \text{ filter}(p) \subseteq \frac{F(\exists) f p}{\exists} \text{ sort } P$ Lemma 8.1, f monotonic on P , p coreflexive	one sorted list built from sorted arguments, instead of a set built and then sorted	
$\langle \text{listcp}(F) \langle g_1, g_2 \rangle \text{ merge } P \text{ thinlist } Q \rangle \text{ minlist } R$ $\subseteq \frac{Q \subseteq P}{\exists} \text{ min } R$ Theorem 8.2	the binary thinning theorem : a fold on sorted lists of partial solutions, thinned at every step	

§14.4. The knapsack problem

(14.4a)

$\text{vol}, \text{wt} : \text{Item} \rightarrow \text{Real}$, $\text{value} \triangleq \text{list}(\text{vol}) \text{ sum}$, $\text{weight} \triangleq \text{list}(\text{wt}) \text{ sum}$.

$\text{subseq} = \langle [\text{nil}, \text{cons}] \cup [\text{nil}, \pi_2] \rangle$, within w the coreflexive on xs with $\text{weight } xs \leq w$.

$\geq \triangleq \leq^\circ$, $R \triangleq \text{value} \geq \text{value}^\circ$, $Q \triangleq R \cap (\text{weight} \leq \text{weight}^\circ)$, $P \triangleq R$.

$FA = 1 + \text{Item} \times A$, $\text{listcp}(F) = \text{wrap} + \text{cpr}$, $g_1 \triangleq \text{list}([\text{nil}, \text{cons}]) \text{ filter}(\text{within } w) = [\text{list}(\text{nil}), h_1]$,
 $g_2 \triangleq \text{list}([\text{nil}, \pi_2]) = [\text{list}(\text{nil}), h_2]$.

$h_1 \triangleq \text{list}(\text{cons}) \text{ filter}(\text{within } w)$, $h_2 \triangleq \text{list}(\pi_2)$.

the law	what it says	(14.4b)
$\frac{\text{subseq (within } w)}{\exists} \text{ min } R$	the specification: a selection of greatest total value that stays within the capacity w	
$\text{subseq (within } w) = \langle ([\text{nil}, \text{cons}] (\text{within } w)) \cup [\text{nil}, \pi_2] \rangle$ fusion, weights non-negative	the selections that fit are themselves a fold, and the fold's algebra already has binary thinning's shape	
$(1 \times R) (\text{cons (within } w)) \subseteq \text{cons (within } w) R$ FALSE	not monotonic on R : a selection of greater value need not still fit once one more item goes in	
$(1 \times Q) (\text{cons (within } w)) \subseteq \text{cons (within } w) Q$ $(1 \times Q) \pi_2 \subseteq \pi_2 Q$	both halves are monotonic on Q once ties in value are broken by weight	
$\langle \text{listcp}(F) \langle g_1, g_2 \rangle \text{ merge } R \text{ thinlist } Q \rangle \text{ minlist } R$ Theorem 8.2 at $P \triangleq R$, F linear	binary thinning, sorting in descending order of value	
$\text{knapsack } w = \langle [\text{nil}, \text{cpr } \langle h_1, h_2 \rangle \text{ merge } R \text{ thinlist } Q] \rangle \text{ minlist } R$	the program; $\text{minlist } R$ becomes head , since packings come out in descending value	

§14.5. The paragraph problem

(14.5a)

```

Line=list+ Word, Para=list+ Line, FA=Word+Word×A, listcp(F)=wrap+cpr.

new (a,xs)=[a]#xs, glue (a,xs)=[a]#head xs#tail xs, partition≐([wrap wrap,newuglue]) : list+
Word→Para.

width≐([length,(length×1) plus succ]), fits w the coreflexive on a line x with width x≤w, ok w the coreflexive
on [x]#xs with width x≤w.

white w x=w-width x, collect≐list(sqr) sum, waste w≐init list(white w) collect.

R≐(waste w)≤(waste w)°, Q≐Rn(head head°), P≐T.

g1≐list([wrap wrap,new]), g2≐list([wrap wrap,glue]) filter(ok w), start≐wrap wrap wrap, h1≐list(new),
h2≐list(glue) filter(ok w).

```

the law	what it says	(14.5b)
$\frac{\text{partition list}^+(\text{fits } w)}{\ni} \min R$	the specification: least waste among the paragraphs whose every line fits	
$\text{partition list}^+(\text{fits } w) = ([\text{wrap wrap}, \text{newu}(\text{glue } (ok\ w))])$ fusion, every word fits on a line by itself	the fitting paragraphs are themselves a fold	
$[\text{wrap wrap}, \text{newu}(\text{glue } (ok\ w))]$ $= [\text{wrap wrap}, \text{new}] \cup ([\text{wrap wrap}, \text{glue}] (ok\ w))$	rewritten into binary thinning's shape $(f_1 p_1) \cup (f_2 p_2)$	
$(1 \times R) \text{ glue} \sqsubseteq \text{glue } R \quad \text{FALSE}$	glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this	
$(1 \times Q) \text{ new} \sqsubseteq \text{new } Q$ $(1 \times Q) (\text{glue } (ok\ w)) \sqsubseteq \text{glue } (ok\ w) Q$ cons monotonic on $\text{collect} \leq \text{collect}^\circ$	both halves are monotonic on Q once ties in waste are broken by the first line	
merge $T = \text{cat}$; $P \triangleq \text{head prefix head}^\circ$ also serves	T needs no sorting at all, and prefix is a linear order on first lines of paragraphs of one input	
$(\text{listcp}(F) \langle g_1, g_2 \rangle \text{ cat thinlist } Q) \min \text{list } R$ Theorem 8.2 at $P \triangleq T$	binary thinning, one candidate kept per first line	
paragraph $w = ([\text{start}, \text{cpr } \langle h_1, h_2 \rangle \text{ cat thinlist } Q]) \min \text{list } R$	the program, g_1 and g_2 split along the coproduct	

§14.6. Bitonic tours

(14.6a)

```

FA=(City×City)+(City×A), the base functor of cons-lists of length at least two; listcp(F)=wrap+cpr; tc :
City×City→Real, neither positive nor symmetric.

start (a,b)=[a,b],[a,b], dropl (a,([b]#xs,ys))=[a]#xs,[a]#ys, dropr (a,(xs,[b]#ys))=[a]#xs,
[a]#ys, tour≐([start,dropl,dropr]).

cost (xs,ys)=outcost xs+incost ys, outcost [a0,...,an]=tc (a0,a1)+...+tc (an-1,an), incost [a0,...,an]=tc
(a1,a0)+...+tc (an,an-1).

next≐tail head, next2≐next×next, R≐cost≤cost°, Q≐Rn(next2 next2°), P≐T, g1≐list([start,dropl]),
g2≐list([start,dropr]).

```

the law	what it says	(14.6b)
$\frac{\text{tour}}{\exists} \min R$	the specification: a least-cost bitonic tour, outward journey and return kept as a pair of lists	
$(1 \times R) \text{ dropl} \sqsubseteq \text{dropl } R \quad \text{FALSE}$ $(1 \times R) \text{ dropr} \sqsubseteq \text{dropr } R \quad \text{FALSE}$	neither drop is monotonic on R : the two edges it adds and removes depend on head and next of both lists	
$(1 \times Q) \text{ dropl} \sqsubseteq \text{dropl } Q$ $(1 \times Q) \text{ dropr} \sqsubseteq \text{dropr } Q$	both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input	
$(\text{listcp}(F) \langle g_1, g_2 \rangle \text{ cat thinlist } Q) \minlist R$ Theorem 8.2 at $P \triangleleft T$, merge $T = \text{cat}$	binary thinning, one candidate per pair of second cities	
$\text{mintour} = ((\text{start wrap, cpr } \langle \text{list}(\text{dropl}), \text{list}(\text{dropr}) \rangle$ $\text{cat thinlist } Q)) \minlist R$	the program: quadratic, because each step adds just two tours to the list kept	

§15. Dynamic Programming

§15.1. Theory

$h : FB \rightarrow B$ a map, $T : FA \rightarrow A$ an F -algebra, $R : B \rightarrow B$.

(15.1a)

$H \triangleq (T)^\circ(h) : A \rightarrow B$, $M \triangleq \frac{H}{\supset} \min R$ the problem to be solved, $(\mu X : G(X))$ the least fixed point of G .

the law	what it says	(15.1b)
$(\mu X : \frac{T^\circ}{\supset} P(F(X)h) \min R) \in M$ Theorem 9.1, h monotonic on R	dynamic programming: decompose in every way, solve each part, keep one optimum per part	
$\frac{T^\circ}{\supset} P(F(M)h) \min R \in M \quad (9.1)$	all Knaster–Tarski leaves to prove	
$\frac{T^\circ}{\supset} P(F(M)h) \min R \in H \quad (9.2)$ $H^\circ \frac{T^\circ}{\supset} P(F(M)h) \min R \in R \quad (9.3)$	(9.1) split by the universal property of $\min$	
$P(X) \min R \in (\exists X) n(\epsilon \setminus (XR^\circ))$ (9.4) = (7.10), (13.1b)	the only fact about $\min$ either proof uses	
$(\mu X : \frac{T^\circ}{\supset} \text{thin } Q P(F(X)h) \min R) \in M$ Theorem 9.2, Q a preorder with $QF(H)h \subseteq F(H)hR$; $\text{thin } Q$ as in (14.1b)	the same with a thinning step: decompositions that can never win are dropped	
the fixed point is unique, and entire Theorem 6.3, T° followed by F 's membership relation inductive; $\frac{T^\circ}{\supset}$ finite and non-empty, R connected	when the recursion can be refined to a recursive function	
$\frac{[V_1, V_2]^\circ}{\supset} \text{thin}(Q_1 + Q_2) P([U_1, U_2]) \min R$ $= (\text{Ran } V_1 \rightarrow W_1, W_2)$ $W_i \triangleq \frac{V_i^\circ}{\supset} \text{thin}(Q_i) P(U_i) \min R$ Proposition 9.1, $V_2 V_1^\circ = \perp$	FA is usually a coproduct, and disjoint ranges split the fixed point into one branch per summand	
$F(R)h \subseteq hR$ $QF(H)h \subseteq F(H)hR$	the two conditions to be checked, in the order the three propositions below discharge them	
$F(R)h \subseteq hR$ Proposition 9.2, $R \triangleq \text{cost} \leq \text{cost}^\circ$, $h \text{ cost} = F(\text{cost})k$, $F(\leq)k \subseteq k \leq$	monotonicity when the cost is itself a fold with a step k monotonic on $\leq$	
$F(Rn(H^\circ H))h \subseteq hR$ Proposition 9.3, $R \triangleq \text{cost} \leq \text{cost}^\circ$, $h \text{ cost} = F(\langle \text{cost}, H^\circ \rangle)k$, $F(\leq \times \mathbb{1})k \subseteq k \leq$, H° simple	monotonicity in context : k may also read the input the part was built from	
$Q \triangleq F(U, V)$ Proposition 9.4, U, V preorders, $F(U, R)h \subseteq hR$, $VH \subseteq HR$	both conditions of Theorem 9.2 at once, split along the two arguments of a bifunctor	

§15.2. The string edit problem

$Op ::= \text{cpy Char} \mid \text{del Char} \mid \text{ins Char}$, $F(A, B) = 1 + (A \times B)$, $\alpha \triangleq [\text{nil}, \text{cons}]$.

(15.2a)

$\text{edit} \triangleq ([\text{base}, \text{step}]) : [Op] \rightarrow [\text{Char}] \times [\text{Char}]$, base returning $([], [])$, $\text{step}(\text{cpy } a, (xs, ys)) = ([a] \# xs, [a] \# ys)$, $\text{step}(\text{del } a, (xs, ys)) = ([a] \# xs, ys)$, $\text{step}(\text{ins } a, (xs, ys)) = (xs, [a] \# ys)$.

$\text{length} \triangleq ([\text{zero}, \pi_2 \text{ succ}])$, $R \triangleq \text{length} \leq \text{length}^\circ$, $U \triangleq \top$, $V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ$, $Q \triangleq 1 + (U \times V)$, empty the coreflexive on (xs, ys) with both lists empty.

unstep implements $\frac{\text{step}^\circ}{\supset} \text{thin}(U \times V)$: $\text{unstep}([a] \# xs, []) = [(\text{del } a, (xs, []))]$, $\text{unstep}([], [b] \# ys) = [(\text{ins } b, ([], ys))]$, $\text{unstep}([a] \# xs, [b] \# ys) = (a \rightarrow [(\text{cpy } a, (xs, ys))], [(\text{del } a, (xs, [b] \# ys)), (\text{ins } b, ([a] \# xs, ys))])$.

the law	what it says	(15.2b)
$\frac{\text{edit}^\circ}{\supset} \min R$	the specification: a shortest edit sequence from which both strings can be reconstituted	
$F(R) \alpha \sqsubseteq \alpha R$ Proposition 9.2 at <code>length</code> , <code>succ</code> monotonic on $\leq$	<code>cons</code> is monotonic on R , so Theorem 9.1 already applies	
$QF(1, \text{edit}^\circ) \alpha \sqsubseteq F(1, \text{edit}^\circ) \alpha R$	the thinning condition sought, over $F(0p, [\text{Char}] \times [\text{Char}])$	
$F(T, R) \alpha \sqsubseteq \alpha R$ left as an exercise	$U \triangleq T$ costs nothing: any two operations may be compared	
$\text{edit}(1 \times \text{suffix}) \sqsubseteq R^\circ \text{edit}$ $\text{edit}(\text{suffix} \times 1) \sqsubseteq R^\circ \text{edit}$	the other half of Proposition 9.4 for $V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ$	
$\text{edit}(1 \times \text{tail}) \sqsubseteq R^\circ \text{edit}$ $\text{edit}(\text{tail} \times 1) \sqsubseteq R^\circ \text{edit}$ $\text{suffix} = \text{tail}^*$, and $BA \sqsubseteq CB \implies BA^* \sqsubseteq C^*B$	one step is enough: drop the operation that produced the head, or weaken its <code>cpy</code> to a <code>del</code> — never lengthening the sequence	
$(\mu X : \frac{[\text{base}, \text{step}]^\circ}{\supset} \text{thin } Q P([\text{nil}, (1 \times X) \text{ cons}]) \min R) \sqsubseteq \frac{\text{edit}^\circ}{\supset} \min R$ Theorem 9.2	a copy, when available, beats a delete or an insert	
$X = (\text{empty} \rightarrow \text{nil}, \frac{\text{step}^\circ}{\supset} \text{thin}(U \times V) P((1 \times X) \text{ cons}) \min R)$ Proposition 9.1	<code>base</code> and <code>step</code> have disjoint ranges	
$\text{mle} = (\text{empty} \rightarrow \text{nil}, \text{unstep list}((1 \times \text{mle}) \text{ cons}) \text{ minlist } R)$	the program: at most two decompositions survive, but the same subproblem is solved many times, so the running time is exponential	
$\text{column } xs \text{ } ys = [\text{mle } (u, ys) \mid u \leftarrow \text{tails } xs]$ $\text{column } xs = ([\text{fstcol } xs, \text{nextcol } xs]), \text{fstcol} = \text{list}(\text{del})$ tails	the tabulation: $\text{mle } (xs, ys)$ needs $\text{mle } (u, v)$ for every tail u of xs and v of ys , so the columns are built right to left	
$\text{column } xs \text{ } ([b] \# ys) = \text{nextcol } xs \text{ } (b, \text{column } xs \text{ } ys)$ $\text{nextcol } xs \text{ } (b, us)$ $= ([\text{base } (b, \text{last } us), \text{step } b]) \text{ xus}$ $\text{xus} = \text{zip } (xs, \text{zip } (\text{init } us, \text{tail } us))$	each column is a fold built bottom to top, over xs zipped with the adjacent pairs of the column to its right	
$\text{base } (b, u) = [[\text{ins } b] \# u]$ $\text{step } b \text{ } ((a, (u, v)), ws) =$ $(a \rightarrow b \rightarrow [[\text{cpy } a] \# v] \# ws,$ $[b \text{ min } R \text{ } ([\text{del } a] \# w, [\text{ins } b] \# u)] \# ws)$ $w = \text{head } ws$; these <code>base</code> , <code>step</code> are not <code>edit</code> 's	an entry depends on the one below it (a delete), the one to its right (an insert), and the one below that (a copy) — quadratic in the two lengths	

§15.3. Optimal bracketing

<code>tree A ::= tip A bin (tree A, tree A), $FX = A + X^2$, so $F(R) = 1 + R^2$; $h \triangleq [\text{tip}, \text{bin}]$, $\text{flatten} \triangleq ([\text{wrap}, \text{cat}]) : \text{tree } A \rightarrow \text{list}^+ A$, $H = \text{flatten}^\circ$.</code> <code>(cost, size) $\triangleq ([\text{opt}, \text{opb}])$, $\text{opt} \triangleq (\text{zero}, \text{st})$, $\text{opb } ((cx, sx), (cy, sy)) = (\text{cb } (sx, sy) + cx + cy, \text{sb } (sx, sy))$.</code> <code>sb</code> associative, so $\text{size} = \text{flatten } sz$ for a map sz; $R \triangleq \text{cost} \leq \text{cost}^\circ$, $g \triangleq [\text{zero}, (1 \times sz)^2 \text{ opb } \pi_1]$, single the coreflexive on singleton lists. <code>splits $\triangleq (\text{inits}^+, \text{tails}^+)$ zip, an implementation of $\frac{\text{cat}^\circ}{\supset}$; <code>array $\triangleq \text{inits list}(\text{row})$, <code>row $\triangleq \text{tails list}(\text{mct})$, <code>col $\triangleq \text{inits list}(\text{mct})$.</code></code></code></code> <code>mix $\triangleq \text{zip list}(\text{bin}) \text{ minlist } R$, <code>next $\triangleq (\pi_1, \text{mix}) \text{ snoc}$, <code>process $\triangleq ((\text{tip wrap}) \times 1) \text{ loop}(\text{next})$.</code></code></code>	(15.3a)
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the law	what it says	(15.3b)
$\frac{\text{flatten}^\circ}{\ni} \min R$	the specification: a least-cost bracketing of $a_1 \oplus \dots \oplus a_n$, the tree whose flattening is the given list	
$[\text{tip}, \text{bin}] \text{ cost} = (1 + (\text{cost}, \text{flatten})^2) \ g \quad (9.5)$	the cost of a node reads only the cost and the flattening of its two subtrees	
$(1 + (\leq \times 1)^2) \ g \in g \leq \quad (9.6)$	g is monotonic on $\leq$ in its two cost arguments	
$F(Rn(\text{flatten } \text{flatten}^\circ))h \in hR$ Proposition 9.3, $H^\circ = \text{flatten}$ a map	monotonicity in context: only trees with the same flattening are compared	
$(\mu X : \frac{[\text{wrap}, \text{cat}]^\circ}{\ni} P([\text{tip}, (X \times X) \text{ bin}]) \min R) \in \frac{\text{flatten}^\circ}{\ni} \min R$ Theorem 9.1	split the list in every way, bracket both halves, join	
$X = (\text{single} \rightarrow \text{wrap}^\circ \text{ tip}, \frac{\text{cat}^\circ}{\ni} P((X \times X) \text{ bin}) \min R)$ Proposition 9.1	<code>wrap</code> and <code>cat</code> have disjoint ranges	
$\text{mct} = (\text{single} \rightarrow \text{head } \text{tip}, \text{splits } \text{list}((\text{mct} \times \text{mct}) \text{ bin}) \min \text{list } R)$	the program; exponential, since the segments of one list overlap	
$\text{mct} = (\text{single} \rightarrow \text{head } \text{tip}, (\text{init } \text{col}, \text{tail } \text{row}) \text{ mix}) \quad (9.7)$	the tabulation: <code>mct xs</code> is needed for every non-empty segment <code>xs</code> , so the values are held as an array of rows	
$\text{col} = (\text{single} \rightarrow \text{head } \text{tip } \text{wrap}, (\text{init } \text{col}, \text{tail } \text{row}) \text{ next}) \quad (9.8)$ $\text{cons } \text{col} = (1 \times \text{array}) \text{ process} \quad (9.9)$ $\text{row} = (\text{single} \rightarrow \text{head } \text{tip } \text{wrap}, (\text{mct}, \text{tail } \text{row}) \text{ cons}) \quad (9.10)$	a column extends the column to its left, a row the row below it; (9.9) is (9.8) rewritten as a loop	
$\text{array} = ([\text{fstcol}, \text{addcol}]), \text{fstcol} \triangleq \text{tip } \text{wrap } \text{wrap}$ $\text{addcol} \triangleq (\pi_1 \text{ tip } \text{wrap}, \text{step}) \text{ cons}$ $\text{step} \triangleq (\text{process } \text{tail}, \pi_2) \text{ zip } \text{list}(\text{cons})$	the program: one fold building the array column by column, cubic in the length of the input	

§15.4. Data compression

<pre> [A] ::= nil snoc ([A], A), list+ A ::= wrap A snoc (list+ A, A), String = [Char], Code ::= sym Char ptr (String, String+), F(Code, String) = 1 + (String × Code), α ≜ [nil, snoc]. decode ≜ ([nil, extend]) : [Code] → String, extend (xs, sym a) = xs # [a], extend (xs, ptr (ys, zs)) = xs # zs when ys # zs is a proper prefix of xs # zs; H = decode°. size ≜ ([zero, distr [1 × c, 1 × p] plus]) with c, p the constant costs of a symbol and a pointer; R ≜ size ≤ size°, Q ≜ F(T + T, prefix°) = 1 + (prefix° × (T + T)), the two T on symbols and on pointers. lrt ws = min(prefix° × (T + T)) {(xs, (ys, zs)) xs # zs = ws, ys # zs a proper prefix of ws}, the longest repeated tail; reduce (ws # [a]) = (zs ≠ [] → [(ws, sym a), (xs, ptr (ys, zs))], [(ws, sym a)]) with (xs, (ys, zs)) = lrt (ws # [a]). </pre>	(15.4a)
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the law	what it says
$\frac{\text{decode}^\circ}{\ni} \min R$	the specification: a smallest code sequence decoding to the given string
$F(R) \alpha \sqsubseteq \alpha R \quad (\tau+\tau)[c,p]=[c,p]$	monotonicity is routine, the two costs being constants
$F(\tau+\tau, R) \alpha \sqsubseteq \alpha R$ decode prefix $\in R^\circ$ decode Proposition 9.4 at $Q \triangleq F(\tau+\tau, \text{prefix}^\circ)$	the thinning condition split in two: pointers are ordered only by how much input they consume, symbols and pointers not at all
decode init $\in R^\circ$ decode	enough for the second: drop the last character of the output by shortening or removing the last code element, never raising the cost
$(\mu X : \frac{[\text{nil}, \text{extend}]^\circ}{\ni} \text{thin } Q P([\text{nil}, (X \times 1) \text{ snoc}]) \min R) \sqsubseteq \frac{\text{decode}^\circ}{\ni} \min R$ Theorem 9.2	between a symbol and a pointer nothing can be decided in advance; between two pointers the longer match wins
$X = (\text{null} \rightarrow \text{nil}, \frac{\text{extend}^\circ}{\ni} \text{thin}(\text{prefix}^\circ \times (\tau+\tau)) P((X \times 1) \text{ snoc}) \min R)$ Proposition 9.1	<code>nil</code> and <code>extend</code> have disjoint ranges
$\frac{\text{extend}^\circ}{\ni} (\text{ws} \# [a]) = \{(\text{ws}, \text{sym } a)\} \cup \{(\text{xs}, \text{ptr } (\text{ys}, \text{zs})) \mid \text{xs} \# \text{zs} = \text{ws} \# [a], \text{ys} \# \text{zs} \text{ a proper prefix of } \text{ws}\}$	the decompositions of one string: take the last character as a symbol, or end with a pointer
reduce implements $\frac{\text{extend}^\circ}{\ni} \text{thin}(\text{prefix}^\circ \times (\tau+\tau))$	thinning leaves at most two: the symbol, and the pointer of <code>lrt</code>
encode = (null $\rightarrow$ nil, reduce list((encode $\times$ 1) snoc) minlist R)	the program, again exponential; the book gives no tabulation for it

§16. Greedy Algorithms

§16.1. Theory

h, T, R, H, M as in (15.1a); additionally Q a **connected** preorder on the sets $\frac{T^\circ}{\supseteq}$ returns, so that $\frac{T^\circ}{\supseteq} \min Q$ is entire. (16.1a)

the law	what it says
$(\mu X : \frac{T^\circ}{\supseteq} \min Q F(X)h) \sqsubseteq M$ Theorem 10.1, the hypotheses of Theorem 9.2	greedy : one decomposition is kept at every step, so P and $\frac{T^\circ}{\supseteq}$ disappear from the recursion (16.1b)
$\frac{[V_1, V_2]^\circ}{\supseteq} \min(Q_1 + Q_2) [U_1, U_2]$ $= (\text{Ran } V_1 \rightarrow W_1, W_2)$ $W_i \triangleq \frac{V_i^\circ}{\supseteq} \min(Q_i) U_i$ Proposition 10.1, $V_2 V_1^\circ = \perp$	Proposition 9.1 with min for thin
$Q \triangleq F(U, V)$ Proposition 9.4	still the way to get both conditions, but $\frac{T^\circ}{\supseteq} \min Q$ must now be entire as well
Theorem 7.2, (13.3e)	that greedy theorem chooses among the results of one relational step of a reduce; Theorem 10.1 chooses among the decompositions of the input, for an arbitrary T rather than an initial algebra, at the cost of h being a map

§16.2. The detab-entab problem

$\text{detab} \triangleq ([\text{nil}, \text{expand}]) : \text{String} \rightarrow \text{String}$ over snoc -lists, $\alpha \triangleq [\text{nil}, \text{snoc}]$, $H = \text{detab}^\circ$; $\text{expand } (xs, a) = (a = \text{TB} \rightarrow \text{fill } xs, xs \# [a])$, $\text{fill } xs = xs \# \text{blanks } (n - (\text{col } xs) \bmod n)$. (16.2a)

$\text{col} \triangleq ([\text{zero}, \text{count}])$, $\text{count } (c, a) = (a = \text{NL} \rightarrow 0, c + 1)$; TB the tab, BL the blank, NL the newline, tab stops every n columns.

$R \triangleq \text{length} \leq \text{length}^\circ$, U the preorder with $a U b \iff a = \text{TB} v a = b$, $V \triangleq \text{prefix}^\circ n (\text{fill } \text{fill}^\circ)$, $Q \triangleq 1 + (V \times U)$.

$\text{unfill } xs$ the shortest prefix of xs with $\text{fill } (\text{unfill } xs) = \text{fill } xs$; tbc the trailing blank count, $\text{triple} \triangleq (\text{unfill } \text{entab}, \text{tbc}, \text{col})$.

the law	what it says
$\frac{\text{detab}^\circ}{\supset} \min R$	the specification: a shortest input detab expands to the given output — detab entab=1 and nothing shorter does
$F(T,R) \alpha \in \alpha R$ left as an exercise	U may be any preorder on characters; $a \ U \ b \iff a=TBva=b$ puts TB below every character, so min prefers a tab to a blank
$\text{detab prefix} \in R^\circ \text{ detab} \quad \text{FALSE}$	at $n=8$, detab [a,b,c,d,e,TB]=[a,b,c,d,e,BL,BL,BL] , whose prefix [a,b,c,d,e,BL,BL] is longer than any input giving it
$\text{detab } V^\circ \in R^\circ \text{ detab} \quad V \triangleq \text{prefix}^\circ n(\text{fill fill}^\circ)$	true once only the prefixes that do not cross a tab stop are allowed
$\text{nil } V^\circ = \text{nil} \quad \text{fill } V^\circ = \text{fill}$ $\text{snoc } V^\circ \subseteq \text{snoc } U(\pi_1 V^\circ)$ left as exercises	the three properties of V the derivation rests on; the second is what fill fill [°] was added for
$\text{expand } V^\circ \subseteq \text{expand } U(\pi_1 V^\circ)$	the claim they add up to: shortening the output either leaves the last step alone or discards it
$\text{detab } V^\circ \subseteq \text{detab } U(\text{init detab } V^\circ)$ <i>init</i> inductive, so the greatest solution is the unique one	hence $\text{detab } V^\circ \subseteq \text{prefix detab}$, and $\text{prefix} \in R^\circ$ finishes Proposition 9.4
$(\mu X : \frac{[\text{nil}, \text{expand}]^\circ}{\supset} \min Q (1+(X \times 1)) [\text{nil}, \text{snoc}]) \subseteq \frac{\text{detab}^\circ}{\supset} \min R$ Theorem 10.1	greedy: one character of input is decided at each step
$X = (\text{null} \rightarrow \text{nil}, \frac{\text{expand}^\circ}{\supset} \min(V \times U) (X \times 1) \text{ snoc})$ Proposition 10.1	nil and expand have disjoint ranges
$\frac{\text{expand}^\circ}{\supset} (xs \# [a]) = \{(ys, TB) \mid \text{fill } ys = xs \# [a]\} \cup \{(xs, a)\}$ the first set is non-empty iff $a=BL$ and $\text{col } (xs \# [a]) \bmod n = 0$	the last output character came from a tab only if it is a blank landing exactly on a tab stop
$\frac{\text{expand}^\circ}{\supset} \min(V \times U) (xs \# [a]) =$ $(a=BL \wedge \text{col } (xs \# [a]) \bmod n = 0 \rightarrow (\text{unfill } xs, TB), (xs, a))$	the greedy step: emit a tab whenever a tab is legal, consuming all the blanks back to the previous tab stop
$\text{entab } xs = \text{entab } (\text{unfill } xs) \# \text{blanks } (\text{tbc } xs)$ (10.1)	what makes triple a snoc-list reduce: the output splits at the last tab stop
$\text{triple} = ([\text{base}, \text{op}])$ and $\text{entab} = \text{triple assocl } \pi_1 (1 \times \text{blanks}) \text{ cat}$	the program: one pass carrying the column and the count of pending blanks
base returns $([], (0, 0))$ op $((xs, (t, c)), a) =$ $(a=BL \wedge (c+1) \bmod n \neq 0 \rightarrow (xs, (t+1, c+1)),$ $a=BL \rightarrow (xs \# [TB], (0, c+1)),$ $a=NL \rightarrow (xs \# \text{blanks } t \# [NL], (0, 0)),$ $(xs \# \text{blanks } t \# [a], (0, c+1)))$	hold a blank back, cash the held blanks in for a tab at a tab stop, flush them at a newline, flush them before anything else

§16.3. The minimum tardiness problem

$FX = 1 + (X \times \text{Job})$, $\alpha \triangleq [\text{nil}, \text{snoc}]$ on schedules, $\beta \triangleq [\text{nil}, \text{snag}]$ on bags, **snag** putting a job into a bag; $\text{bagify} \triangleq ([\beta]) : [\text{Job}] \rightarrow \text{Bag Job}$, $H = \text{bagify}^\circ$.

$\text{ct}, \text{dt}, \text{wt} : \text{Job} \rightarrow \text{Real}$ the completion, due and weighting quantities of a job; $\text{penalty } (xs, j) = (\text{sum } (\text{list}(\text{ct}) xs) + \text{ct } j - \text{dt } j) \times \text{wt } j$.

$\text{cost} \triangleq \frac{\text{prefix}}{\supset} P(\alpha^\circ [\text{zero}, \text{penalty}]) \text{ max} \leq$, $\text{cost } [] = 0$, $\text{cost } (xs \# [j]) = \text{bmax } (\text{cost } xs, \text{penalty } (xs, j))$, $R \triangleq \text{cost} \leq \text{cost}^\circ$.

$\text{perm} \triangleq \text{bagify bagify}^\circ = ([\text{nil}, \text{add}])$, $\text{add } (xs, j) = ys \# [j] \# zs$ for some $xs = ys \# zs$.

$k \triangleq [\text{zero}, \text{assocr } (1 \times ((\text{bagify}^\circ \times 1) \text{ penalty})) \text{ bmax}]$, $f \triangleq [\text{zero}, (\text{bagify}^\circ \times 1) \text{ penalty}]$, $Q \triangleq f \leq f^\circ$.

the law	what it says	(16.3b)
$\frac{\text{bagify}^\circ}{\supset} \min R$	the specification: an ordering of the given bag of jobs with least maximum penalty	
$F(Rn(\text{bagify } \text{bagify}^\circ))\alpha \sqsubseteq \alpha R \quad (10.2)$	monotonicity in context — only schedules of one bag are compared	
$(Qn(\beta\beta^\circ))F(\text{bagify}^\circ)\alpha \sqsubseteq F(\text{bagify}^\circ)\alpha R \quad (10.3)$	the greedy condition, also in context	
$\text{cost } (xs\# [j]) = \text{bmax } (\text{cost } xs, \text{penalty } (\text{perm } xs, j))$ $\alpha \text{ cost} = F((\text{cost}, \text{bagify}))k, F(\leq \times 1)k \sqsubseteq k \leq$ Proposition 9.3	(10.2): penalty sums the completion times of xs , and a sum does not see the order	
$\alpha \text{ cost} = (g, h) \text{ bmax} \quad (10.4)$ $g \triangleq [\text{zero}, \text{penalty}] \quad (10.5)$ $h \triangleq [\text{zero}, \pi_1 \text{ cost}] \quad (10.6)$	the cost of a schedule split into the penalty of its last job and the cost of the rest	
$\text{add} \sqsubseteq \pi_1 R \quad (10.7)$ $\beta \text{ bagify}^\circ = F(\text{bagify}^\circ) [\text{nil}, \text{add}] \quad (10.8)$	adding a job never lowers the cost; and bagify [°] is itself a reduce on bags. Together: $\beta \text{ bagify}^\circ \sqsubseteq F(\text{bagify}^\circ)h \leq \text{cost}^\circ$	
$F(\text{bagify})Q^\circ F(\text{bagify}^\circ) = g \geq g^\circ$, met by $Q \triangleq f \leq f^\circ, f \triangleq [\text{zero}, (\text{bagify}^\circ \times 1) \text{ penalty}]$	the specification Q has to satisfy, and the choice that meets it: a Q -minimum is a job of least penalty	
no greedy reduce exists	a greedy snoc-list reduce would also solve every prefix of the input, and the best schedule of a prefix need not extend to a best schedule of the whole	
$X = (\text{null} \rightarrow \text{nil}, \frac{\text{snag}^\circ}{\supset} \min Q (X \times 1) \text{ snoc}) \quad \text{Theorem 10.1, Proposition 10.1}$	nil and snag have disjoint ranges	
$\text{schedule} = (\text{null} \rightarrow \text{nil}, \text{pick } (\text{schedule} \times 1) \text{ snoc})$ $\text{pick} \sqsubseteq \frac{\text{snag}^\circ}{\supset} \min Q$, a partial function	the program: repeatedly remove a job of least penalty and put it last; quadratic in the number of jobs	

§16.4. The TeX problem

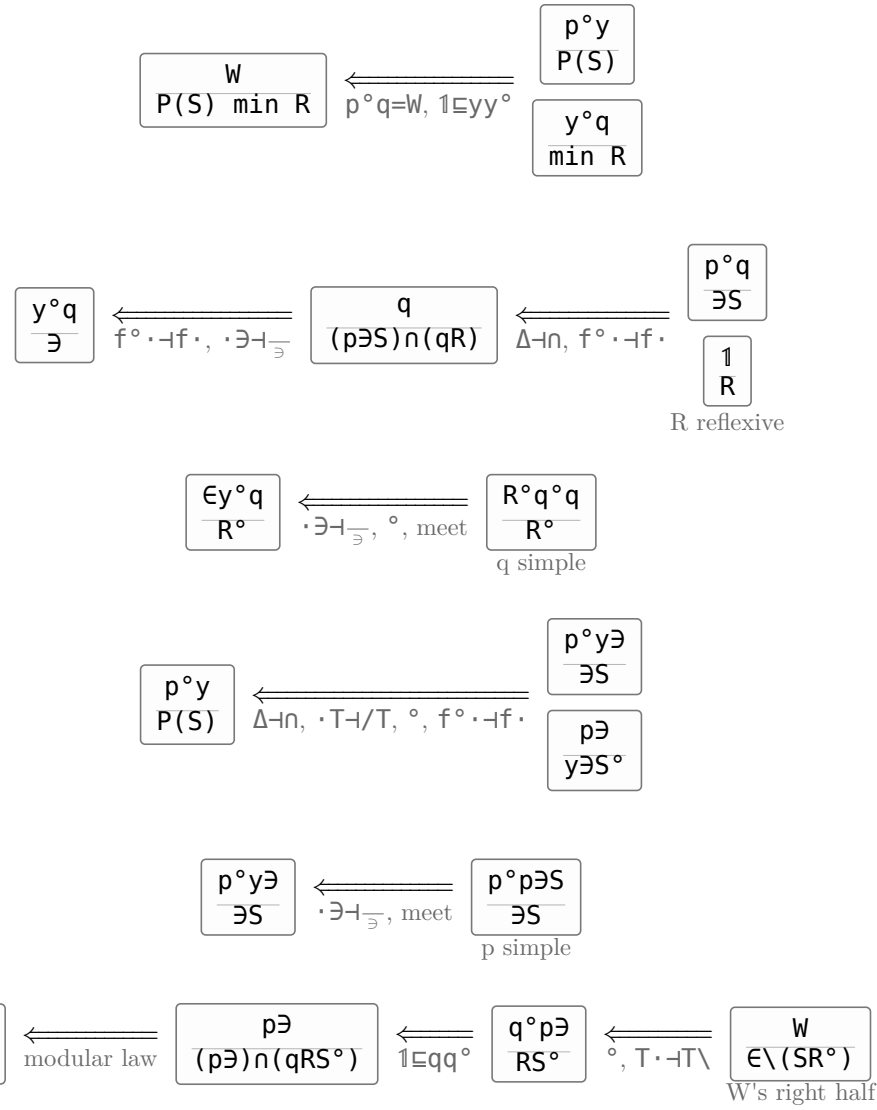
$\text{intern} \triangleq \text{val round} : \text{Decimal} \rightarrow [0, 2^{16}), \text{val} \triangleq ([\text{zero}, \text{shift}]), \text{shift } (d, r) = (d+r)/10, \text{round } r \text{ rounds } 2^{16}r \text{ to the nearest integer: } \text{round } r = n \iff 2n-1 < 2^{17}r < 2n+1.$ $\text{interval } n = ((2n-1)/2^{17}, (2n+1)/2^{17}), r \text{ inrange } (a, b) \iff a < r < b, \text{round}^\circ = \text{interval inrange}, R \triangleq \text{length} \leq \text{length}^\circ.$ Interval the pairs (a, b) with $0 < b < 1$ and $a < b$ (10.9) $[\text{arb}, \text{step}] : 1 + (\text{Digit} \times \text{Interval}) \rightarrow \text{Interval}$, $\text{step } (d, (a, b)) = ((d+a)/10, (d+b)/10).$ $FX = 1 + (\text{Digit} \times X), \alpha \triangleq [\text{nil}, \text{cons}], h \triangleq ([\text{arb}, \text{step}]) = H^\circ, ! : \text{Digit} \times \text{Interval} \rightarrow 1, Q \triangleq (l^\circ !^\circ r) \cup 1, w \triangleq 2^{17}.$	(16.4a)
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the law	what it says
$\frac{\text{intern}^\circ}{\ni} \min R = \text{interval } \frac{\text{inrange val}^\circ}{\ni} \min R$	the specification: a shortest decimal whose internal representation is the given multiple of 2^{-16} . round° is not a map, but interval is, so it comes out of the $\ni$
$\text{val inrange}^\circ = ([\text{arb}, \text{step}])$ fusion; $\text{zero inrange}^\circ = \text{arb}$, $\text{shift inrange}^\circ = (1 \times \text{inrange}^\circ)$ step	the converse of val , cut down to intervals, is a reduce on cons-lists
$F(R) \alpha \sqsubseteq \alpha R$	cons is monotonic on R — routine
$\alpha^\circ F(h) Q^\circ \sqsubseteq R^\circ \alpha^\circ F(h)$	the greedy condition of Theorem 10.1, converted
$\frac{\text{arb}^\circ}{\ni} (a, b) = (a < 0 \rightarrow \{*\}, \{\})$ $\frac{\text{step}^\circ}{\ni} (a, b) = \{(d, (10a-d, 10b-d)) \mid 0 < 10b-d < 1\}$	the decompositions of an interval: stop, or take one more digit
$0 < 10b-d_1 < 1$ and $0 < 10b-d_2 < 1$ imply $d_1 = d_2$	step° is in fact a map, d the digit with $0 < 10b-d < 1$, so $\frac{[\text{arb}, \text{step}]^\circ}{\ni}$ returns at most two elements and Q need only choose between them
$Q \triangleq (\text{!}^\circ \text{!}^\circ r) \cup \text{!}$, and $\text{! nil} \sqsubseteq \text{cons } R^\circ$ from $\text{! nil length} \sqsubseteq \text{cons length} \geq$	stop whenever stopping is legal: the empty decimal is shorter than any other
$(\mu X : \frac{[\text{arb}, \text{step}]^\circ}{\ni} \min Q F(X) \alpha) \sqsubseteq \frac{\text{intern}^\circ}{\ni} \min R$ Theorem 10.1	greedy: emit the one digit the interval allows, until the interval contains zero
$\text{extern} = \text{interval } f$ $f(a, b) = (a < 0 \rightarrow [], [d] \# f(10a-d, 10b-d))$	the program, with d the digit above
$\text{extern } n = f(2n-1, 2n+1)$ $f(p, q) = (p \leq 0 \rightarrow [], [d] \# f(10p-w d, 10q-w d))$ $d = (10q) \text{ div } w, w = 2^{17}$	the same in integer arithmetic only, as chapter 3 required of intern : every interval reached is $(p/w, q/w)$

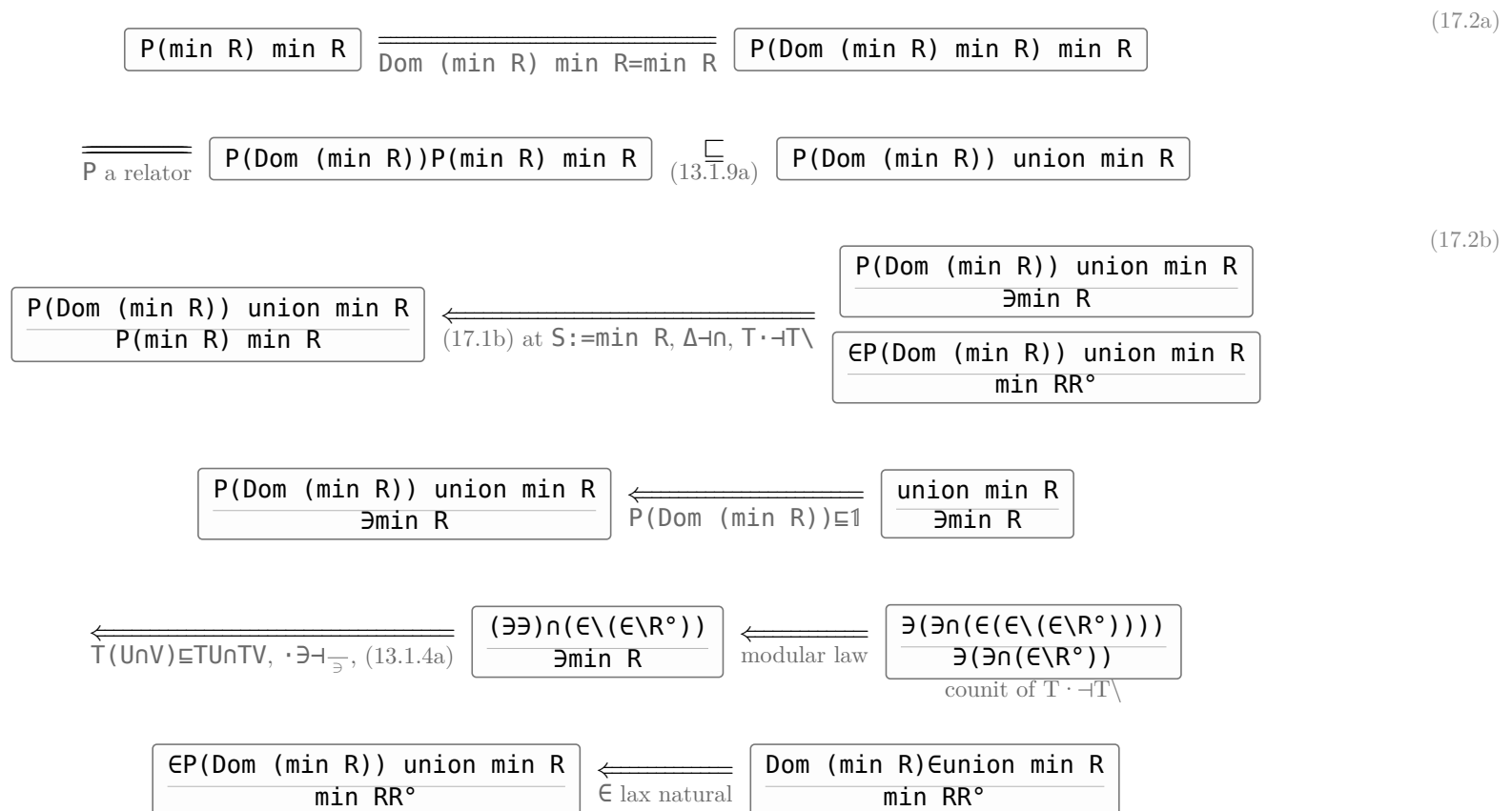
§17. Appendix

§17.1. $P(S) \min R = (\exists S) \cap (\epsilon \setminus (SR^\circ))$

(p, q) tabulates $W \triangleq (\exists S) \cap (\epsilon \setminus (SR^\circ))$, and $y \triangleq \frac{(p \exists S) \cap (qR)}{\exists}$, a map.



§17.2. $P(\min R) \min R = P(\text{Dom}(\min R)) \cup \min R$



$$\begin{array}{ccc}
 \overleftarrow{\text{Dom } (\min R) \sqsubseteq \min R (\min R)^\circ} & \boxed{\frac{\min R (\min R)^\circ \in \text{Union } \min R}{\min R R^\circ}} & \overleftarrow{(\min R)^\circ \sqsubseteq \epsilon, \epsilon \in \text{Union } \epsilon} \boxed{\frac{\min R \min R}{\min R R^\circ}} \\
 & & \text{UP of min}
 \end{array}$$